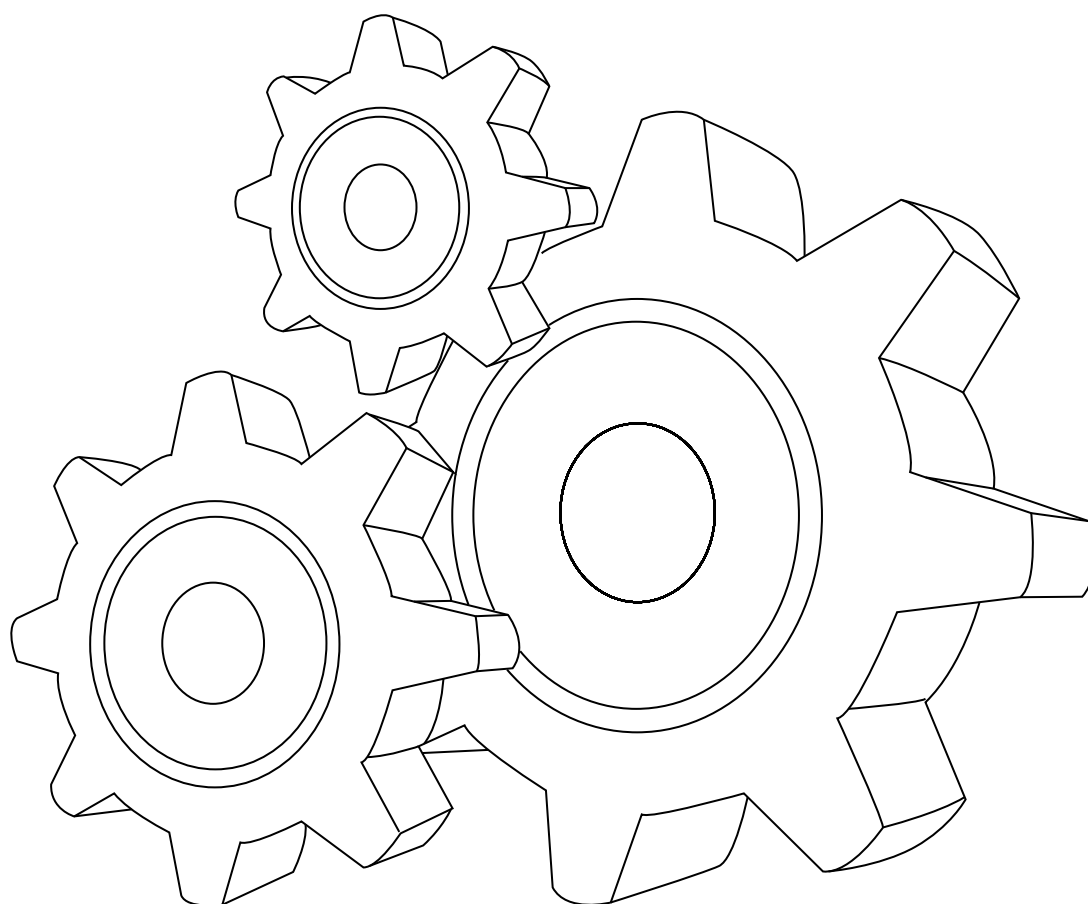




SQS250CSA Operation and Maintenance Manual



XCMG Xuzhou Truck-mounted Crane Co. , Ltd



Foreword

Thank you for purchasing and for using the SQS250CSA loader crane produced by our company.

This product has two functions: lifting and transportation. With the advantages of compact structure, easy to operate, safety and convenience, it is suitable for transportation, docks, warehouse and other places, as well as cargo loading and unloading and long-distance transfer, improve efficiency and save the time and labour.

To ensure your safety and safe operation, you should read this manual carefully to become sufficiently familiar with the procedure for operation, inspection and maintenance of this machine before operation . Truck with loading crane driver, crane operator and crane maintenance man should receive training and become familiar with the heavy duty truck and crane operating procedures.

For prolonging the service life of the truck, please comply with the regulations and requirements of truck operating and maintenance.

The operation and maintenance and notices in the manual are only used for loader crane, if you want to perform operating and maintenance work on the truck, be sure to refer to the truck operation and maintenance manual.

CAUTION

- 1. The person who has not received training is not allowed to operate the machine.**
- 2. Make sure that do not refit the machine or replace its parts with non-original without authorization and permission of manufacturer, and therefore manufacturer has no responsibility for the adverse effect arising from the reasons .**
- 3. All the specification, annex, operation and maintenance are subject to technical modification without notice.**



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Chapter 1 General description

1.1 Definition of glossary

The following is glossary of terms of this manual, with brief explanation.

Controllable safety device: devices controlled by operator to ensure safety.

Articulation point: articulated joint between two or more elements.

hook: attach the end of boom for lifting goods.

Hydraulic telescopic boom: a boom controlled by oil cylinder, can carry out telescopic movement.

Limit point: it can't be movement if exceed the point.

Moment: the product of load applied and its distance from slewing axis. (T.M)

Operator: person who operate the crane.

Nameplate of lifting graph: device for express the 90%-100% lift capacity of crane.

Working range: horizontal distance from center of hook to slewing center.

Passive safety: devices can't be controlled to ensure safety.

Power system: hydraulic drive system, used to realize various movements of crane.

Rated lift capacity: maximum lift capacity at given distance.

Standard device: crane without optional attachment.

1.2 Symbols

In this manual you will come across various symbols accompanied by a written message. The symbols are intended to alert you of the specific situations which are described in the message.

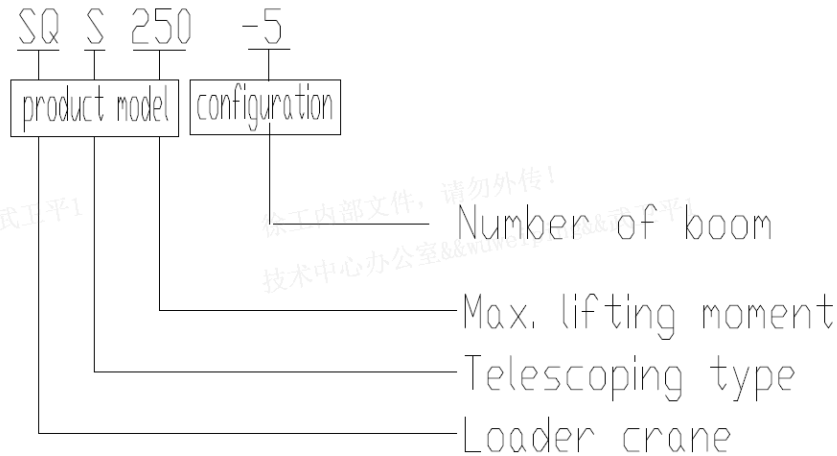
There are three main symbols as follows :

Warning : give the operator of information on methods or procedures that are either prohibited or recommended.

Danger : warn the operator or person of the danger condition around.

Caution : give the operator of the conditions that may be occurred and damage to structure or goods.

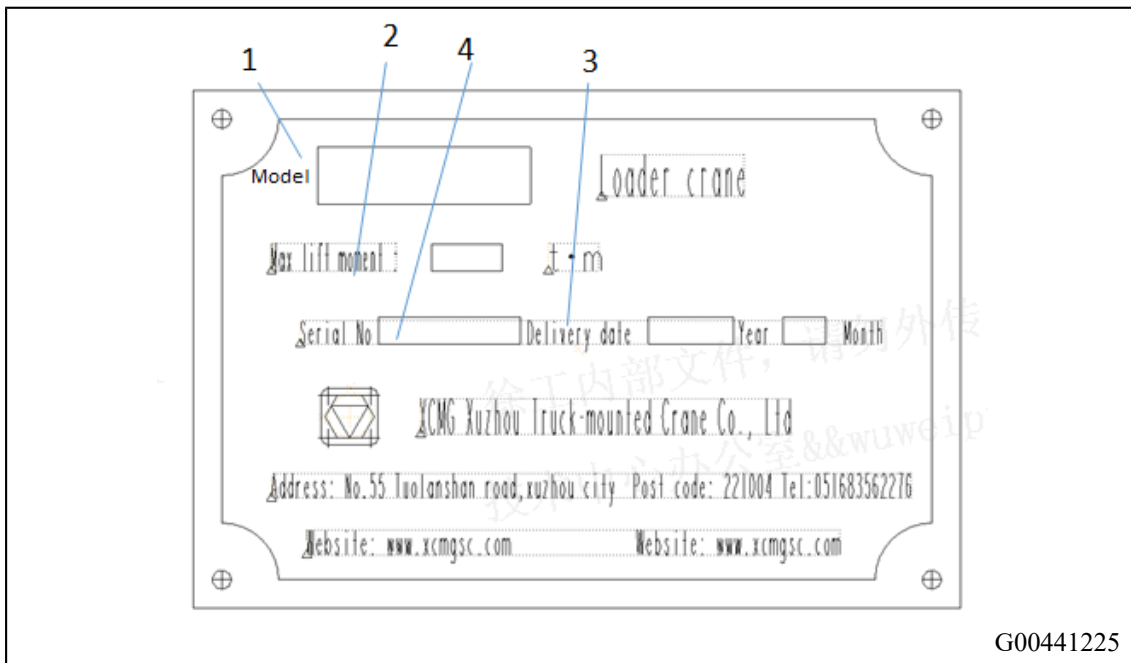
1.3 Crane model and meaning



1.4 Loader crane nameplate

Crane nameplate is installed on the column,
the nameplate includes following contents :

1. Crane model
2. Max. lift moment
3. Delivery date
4. Delivery No.



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1.5 Warning on the crane

The below is the list of points need to attention :

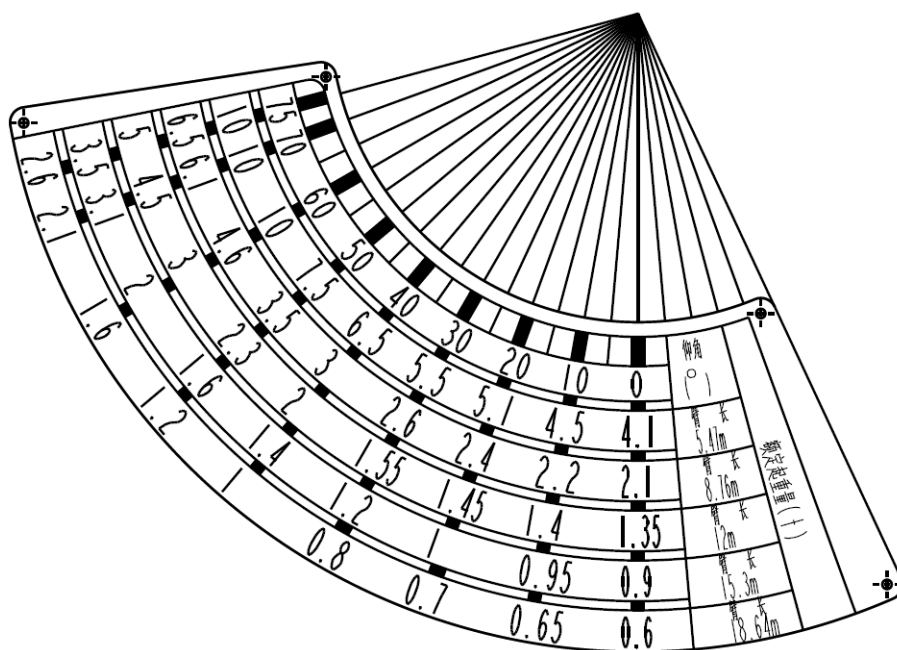
The indicator indicates the following hazards: do not stay within the crane working range.



Do not stand under boom

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This range indicator shows that the maximum lifting capacity with boom in various telescoping state and elevation angle, it is not allowed to lift the load beyond the value specified by the range indicator.

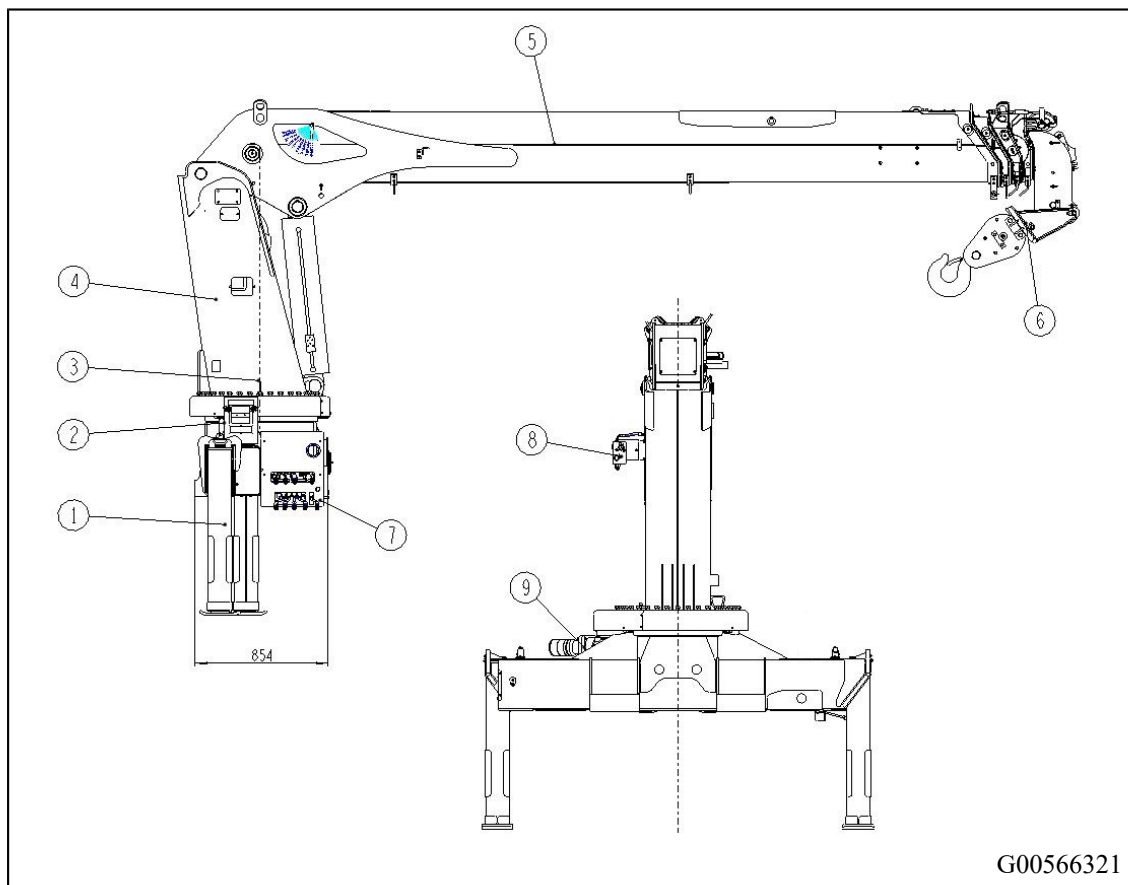


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1.6 Main part name of loader crane

Please refer to the figure below which may help you to recognize the name of components designated in this manual.

1. stabilizer extension assembly 2. Base assembly 3. slewing distributor
4. column assembly 5. boom assembly 6. hook
7. control system 8. lifting mechanism 9. slewing mechanism assembly



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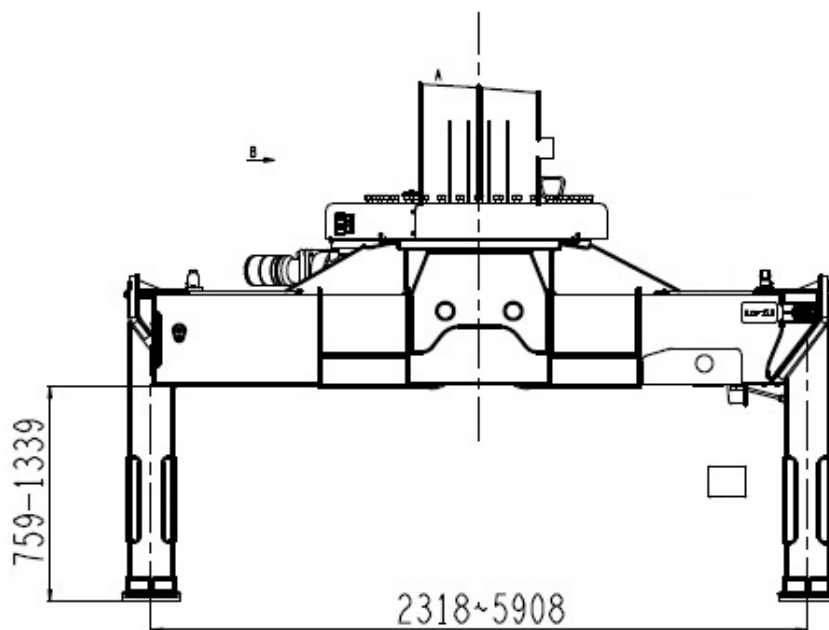
Chapter 2 Description of crane performance data

2.1 Working parameters

The table below lists working parameters of loader crane when leaves the factory.

Category	Item	Unit	SQS250CSA
Lifting performance parameter	Max. rated lift capacity	t.m	25
	Min. working range	m	2.5
	Max. working range	m	18.2
	Max. lifting height	m	20
	Boom length	m	5.47~18.64
	Rotation angle	°	Full rotation
	Luffing angle	°	0~75°
	Max. rated lifting capacity	kg	10000
	Installation space	m	1.2
	Rated pressure of hydraulic system	MPa	26
	Max. flow of hydraulic system	L/min	63+50
	Crane weight	kg	4100

2.2 Slewing base and stabilizer extensions



2.3 Hydraulic system parameter

Item	Unit	Value
Oil pump displacement	ml/r	63+50
System flow	l/min	63+50
Max. working pressure	MPa	26
Power recommended	KW	49
Ambient temperature	OC	-25 °C-+40 °C
Hydraulic pump: recommended gear pump	Model: SGP2063/2050	(can supply according to the demands of user)
Hydraulic oil tank capacity	L	200

2.4 Movement speed

The following table shows the working time and technical parameters of crane.

Category	Item		Unit	Value
Working speed parameters	Boom extensions speed	Full range (extend)	s	≤40

Category	Item		Unit	Value
		Full range (retract)	s	≤ 22
	Boom luffing speed	Full range (up)	s	≤ 25
		Full range (down)	s	≤ 15
	Vertical leg cylinder speed	Full range (extend)	s	≤ 10
		Full range (retract)	s	≤ 8
	Horizontal leg cylinder speed	Full range (extend)	s	≤ 12
		Full range (retract)	s	≤ 10
	Rotation speed		r/min	≤ 3
	Rotation range		°	Continuous slew
	Lifting speed of single rope (4 part lines)		m/min	≥ 16

⚠ CAUTION

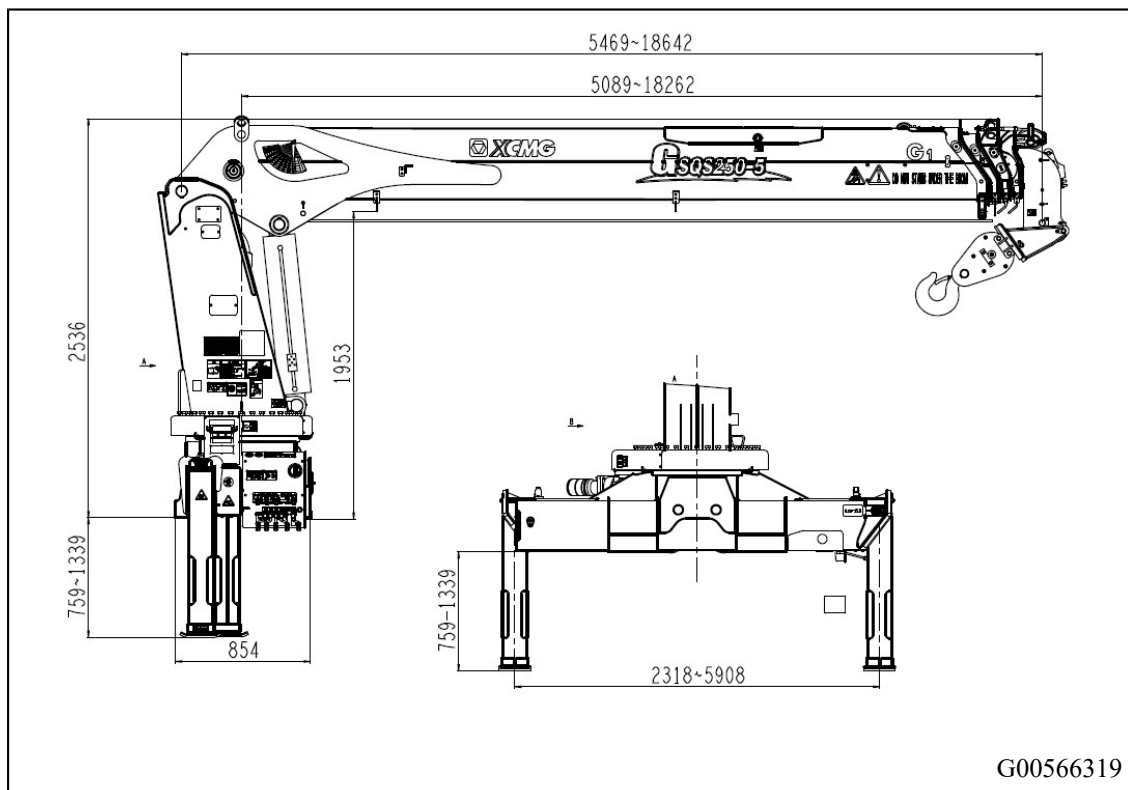
1. The values given in the table refer to the theory time under the conditions of crane movements without load on hook, and pump is at the rated flow.
2. With regard to slewing, the rotation time refers to the time required for the crane to rotate of 360°(one turn).

2.5 Overall dimensions

The figure below shows the minimum dimensions of SQS250CSA crane in fully retracted state.

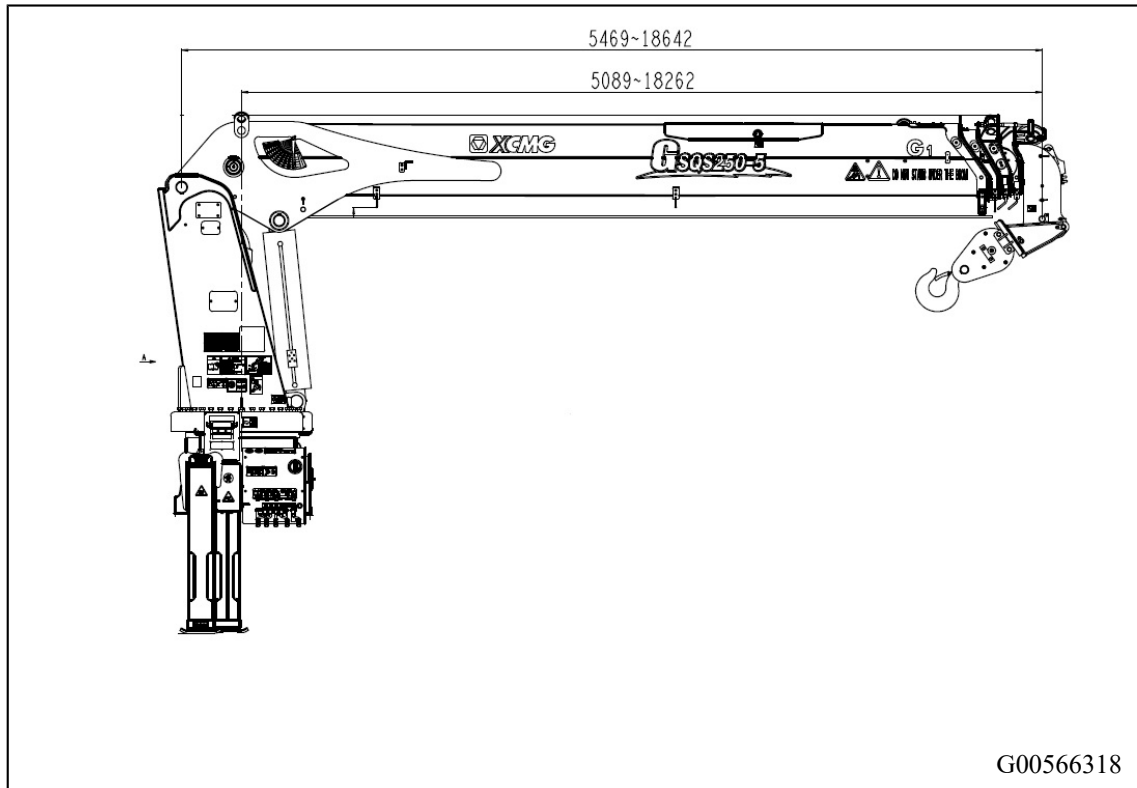
⚠ CAUTION

To drive safely on the highway, the combination of crane and chassis shall meet the requirements of traffic & related law and rules.

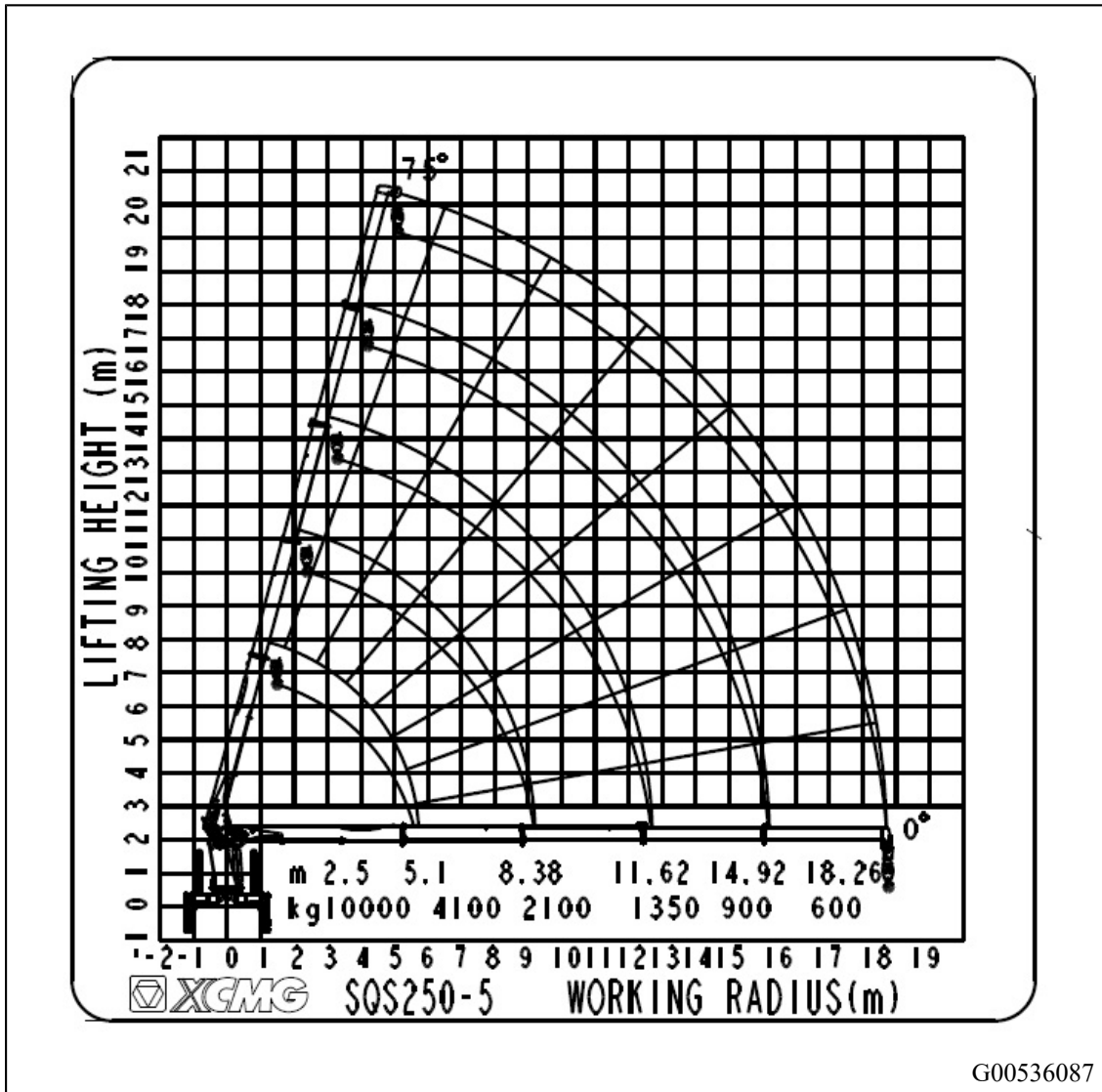


2.6 Boom working range

The figure below shows the maximum & minimum dimensions of SQS250CSA crane in full-extended and full-retracted status in horizontal position.

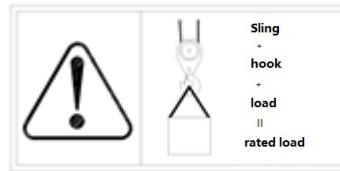


2.7 Crane lifting performance table



CAUTION

1. The maximum total rated load in the figure above is based on the conditions that set the outriggers on the firm and level ground, including the mass of hook, pulley (60kg)and wire rope .
2. The working radius in the figure above refer to the actual radius with the affect of flexibility , so the measured actual radius determines the allowable load value.
3. The load value in the figure above is reduced by 25% when the crane is operated within the range of 180° with the boom is in front(in the travelling direction of vehicle).
4. The crane' s performance characteristics will decrease as a result of combination with a not entirely suitable vehicle, the parameters given in the figure will no longer be useful, several values may not correspond to the data given below, similarly, lifting performance parameters calibrated at other locations on the crane will also be decreased.

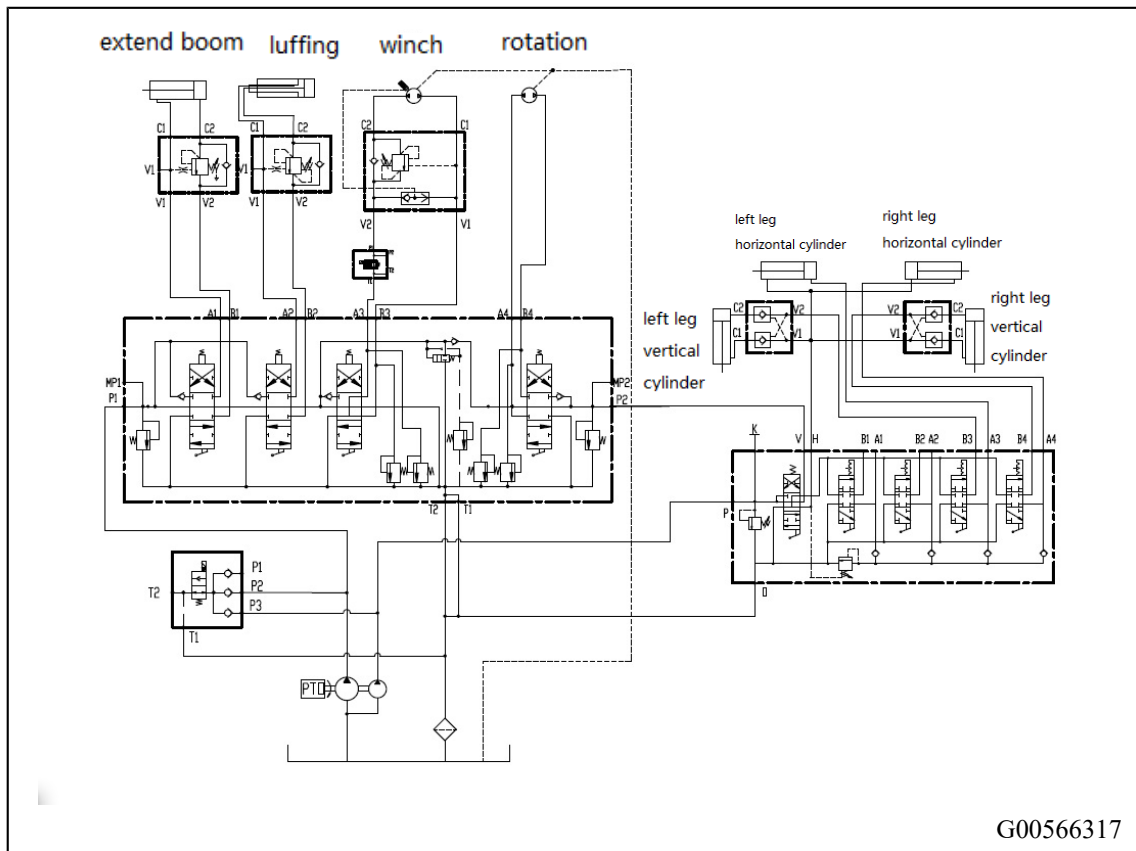


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CAUTION

The load on the label is the maximum permissible total mass(contains hook and slings) for operation at different working range.

2.8 Hydraulic system schematic diagram

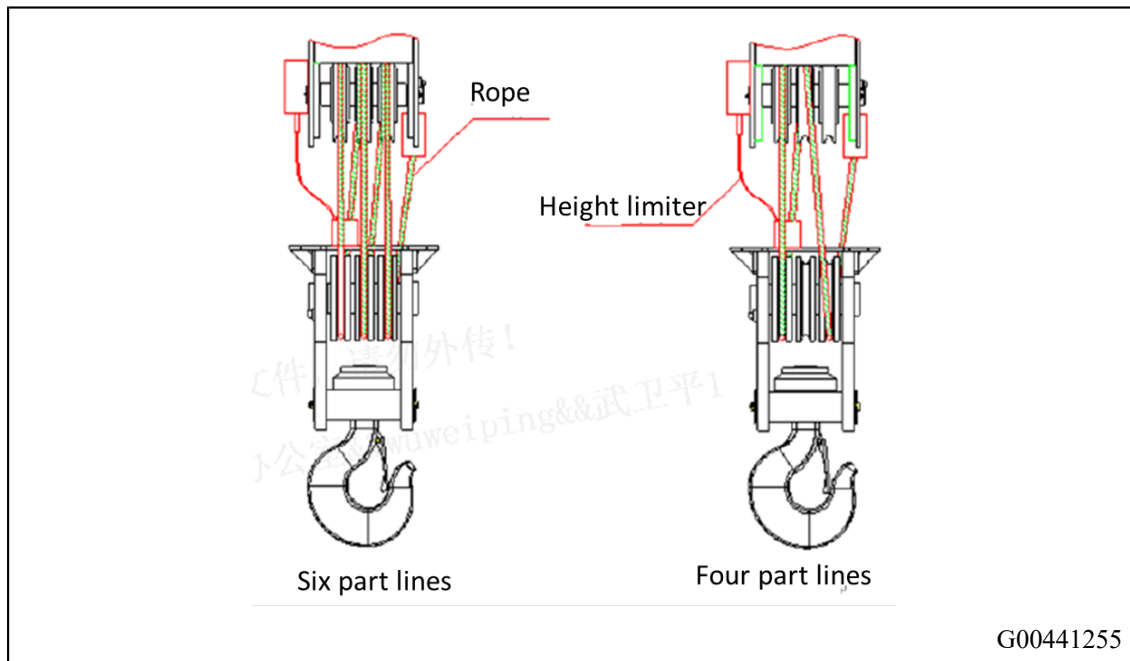


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2.9 Part line selection of wire rope

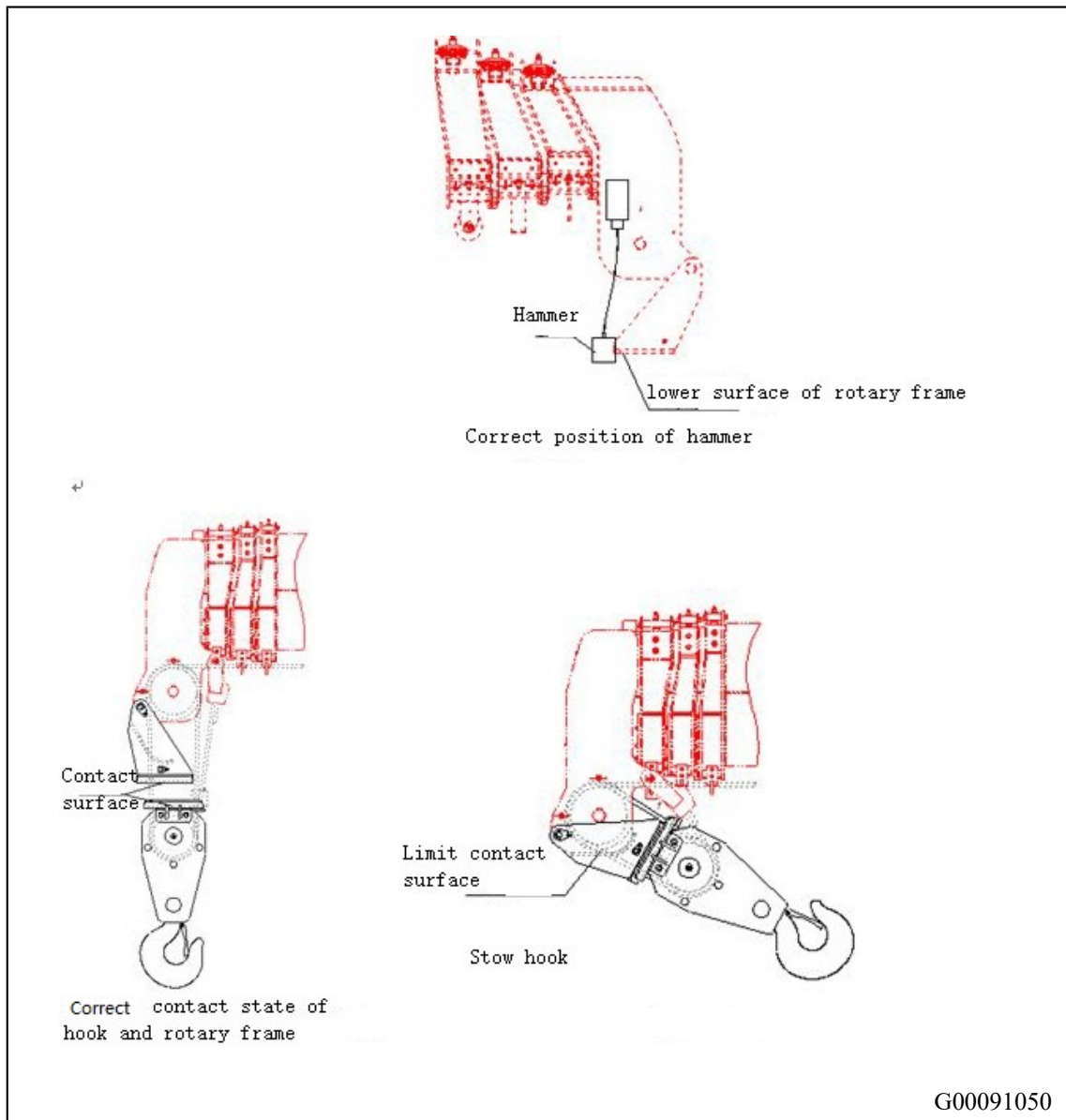
Caution! The selection of wire rope shall meet the requirements when loader crane lifting goods.

1. When the crane leaves the factory, the hook wire rope adopts four part lines, which can meet the working condition that the boom length is greater than 5.7 m, and can improve the working efficiency.
2. When the boom length is less than 5.7 m and the lifting capacity is more than 9 tons, need to replace the part lines of wire rope with 6 part lines of that.



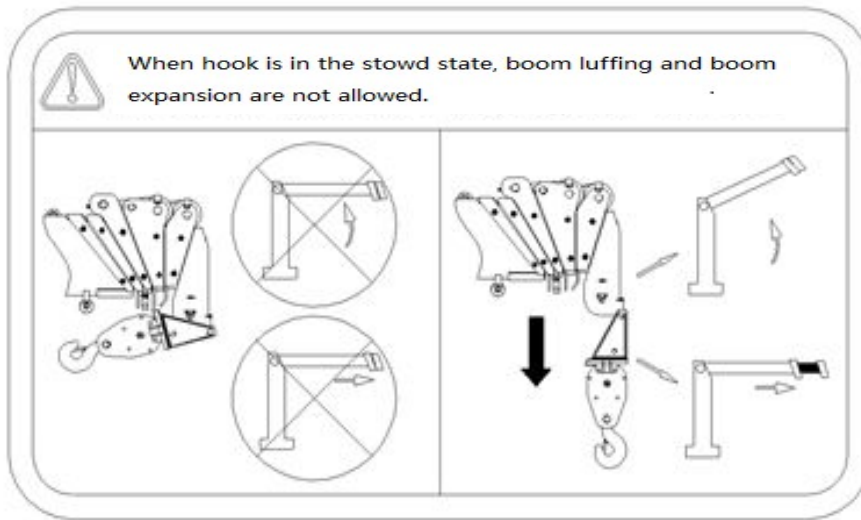
2.10 Automatic stow hook

1. Check if the winding mode of wire rope corresponding to the selected part lines is correct(as shown in the figure above) .
2. Confirm the hammer position of height limiting device: when the rotary frame is horizontal, the lower plane of the rotary frame is between the upper and lower parts of hammer.
3. Operate handle to lift hook, stop lifting and lower the hook when the top plate of hook will be in contact with the lower plane of rotary frame under the boom head, the two sides are about 3 cm apart, make sure that the top plate is level or slightly lower along the crane forward(otherwise, repeat the action)
4. When hook and rotary frame have no obvious swing, then operate handle to tighten the wire rope at a half of the no-load lifting speed, hook contacts the rotary frame and rotate together, when approaching or moving to the limit position, stow and fix it.



⚠ CAUTION

1. The hook can only be stowed when the contact surface is completely in contact, to avoid damage to wire rope and other parts.
2. After the hook is stowed by automatic hook stowing mechanism, the crane is not allowed to do any action such as boom luffing and extension.(see below figure)
3. After stowing hook and travelling, stop vehicle and check if the hook is stowed in place(because the wire rope is flexible), make sure that the stowing function is in the best working condition and avoid damage to wire rope or affect other functions.





Chapter 3 Common fault analysis and elimination

3.1 Common fault analysis and elimination

Faults	Cause	Elimination
Extension cylinder vibrates, boom extension crawls	1) Air in the hydraulic system 2) Seal in the cylinder is aging 3) There is dirt in the balance valve 4) Boom lack of lubricating oil	1) Remove air from hydraulic system 2) Replace cylinder seals 3) Clean balance valve 4) Add lubricating oil
Working speed is still too low without load	1) Rushed or dented suction hose 2) Air enters through suction pipe	1) Replace suction pipe 2) Tighten the connector of suction pipe
Boom extensions can't be extended in sequence	1) Lack of lubricating oil 2) Sliding block is damaged 3) Improper adjustment for telescopic boom valve	1) Add lubricating oil 2) Replace sliding block 3) Adjust telescopic boom valve
Crane can't lift rated load	1) Hydraulic pump power is insufficient 2) Wrong relief valve settings 3) Hydraulic pump seals are damaged	1) Replace hydraulic pump 2) Adjust relief valve pressure again 3) Replace hydraulic pump seals
Boom falls automatically after lifting load	1) Worn piston rod seals of derricking cylinder 2) Balance valve port were jammed or the reset spring is damaged.	1) Replace cylinder seals 2) Clean balance valve and replace spring
Crane can't rotate correctly	1) Truck exceed the maximum allowable gradient. 2) Foreign matters in the buffer valve 4) Worn the support sleeve inside the gear pole 5) No power in the system	1) Adjust the truck to the allowable error 2) Clean or replace the slewing buffer valve 4) Replace support sleeve 5) Check the PTO joint
Articulation point or slewing parts sounds	Lack of lubricating oil	Greased it at specified intervals

Faults	Cause	Elimination
Outrigger leg cylinder can't support the load	1) Two-way hydraulic lock failure 2) The piston seals of leg cylinder are damaged	1) Clean or replace valve lock 2) Replace seals
Cylinder leak oil	1) End cover seals are aging and damaged 2) The piston seals wear	Replace seals
Noise is big, pressure fluctuates greatly, hydraulic valve outcry	1) Suction hose or suction net is jammed 2) Oil viscosity is too high 3) Bad seal for suction port and air is entered 4) The parts in the pump were worn 5) The system pressure is too high	1) Remove the dirt 2) Replace hydraulic oil as required or warm it 3) Replace seals and tighten the screws 4) Replace or repair inner parts 5) Adjust the system pressure again
Clearance jerk was shown when winch is lifting or falling	1) The rope is twisted or disordered on the drum 2) Brake friction plate is damaged or friction area is less than 80% 3) Dirt in the lifting motor 4) Oil supply is insufficient and the system pressure is low	1) Put down wire rope, remove stress and rewind the rope 2) Replace friction plate and adjust related parts, make sure that the friction area is more than 80% 3) Clean and remove dirt 4) Check if oil pump work normally, adjust hydraulic oil temperature, keep the oil level suitable, adjust relief valve pressure value when it is lifting load, and replace spring if the pressure is not enough



Chapter 4 Maintenance

4.1 Notices before maintenance

The following maintenance marked * must be carried out by manufacturer.

Maintenance jobs must be performed with the truck engine off and the key operate switch disengaged.

Before carry out work on the pressure lines, release the pressure first(with the truck engine off) by operating the crane control levers in both direction.

Keep all handles, footplate and worktable free of dirt, adding antiskid agent to prevent sliding down.

Don't spray water on electrical system element (for example: sensor, wire connector) directly. If water enters electrical system, it may cause malfunction. be sure to protect all the electric components and joints when cleaning crane.

Nevertheless, we recommend you make periodic inspection to check that the protection treatment is in good condition or not, and treat it again if necessary.

Before starting, make sure that there are not tools, rag, and any other things leave on the movement parts.

4.2 Maintenance schedule

The following contents are the main inspection items of crane within the specified period.

Maintenance per **50** hours

- _ Check the leaks for system connections.
- _ Check the leaks for cylinders
- _ Check the tightness of mounting bolts and other tighteners.

Maintenance per **100** hours or 1 month

- _ Apply grease to transmission shaft splines and universal joint.
- _ Grease telescoping boom.

Maintenance every **450** hours or every **6** months.

- _ Grease slewing unit of column.
- _ Grease articulation points.
- _ Grease the piston rod out of cylinder.
- _ Check the wear for sliding pads , replace it if wear.
- _ Replace hydraulic oil filter and air breather filter.
- _ Check the wear for wire rope and replace it in time.

Maintenance every **900** hours or once a year

- _ Check hydraulic oil cylinder (*)
- _ Check tightness of base screw.
- _ Check hydraulic system settings and efficiency of safety devices. (*)
- _ Check the tightness of crane mounting bolts.
- _ Check crane steel structure (*)
- _ Check/replace adjusting screws and sliding block.

_Replace hydraulic oil

If the crane remains idle more than 6 months, you should take some measure as follows.

_Keep the body free of oil, grease and dirt.

_Retract all the piston rod of cylinder completely

_Grease movement parts.

_Keep the rope free of dust and grease it with ZG-S calcium base grease again.

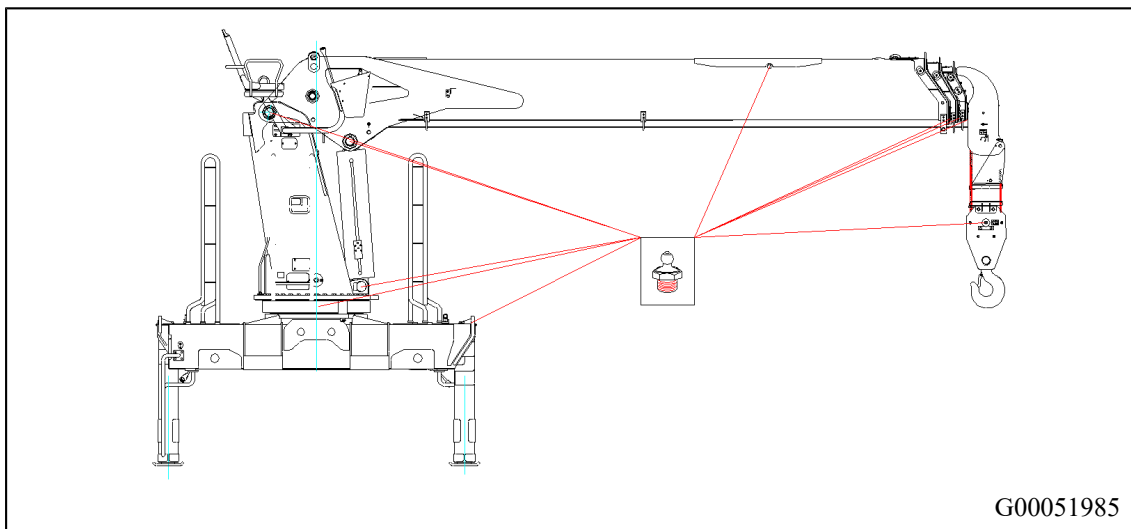
_Generally stored in the ventilated dry warehouse, it should be covered by waterproof cloth if it is stored in the open air.

_Start it once a month and idling the structure to check if it is in good condition.

4.3 Crane maintenance

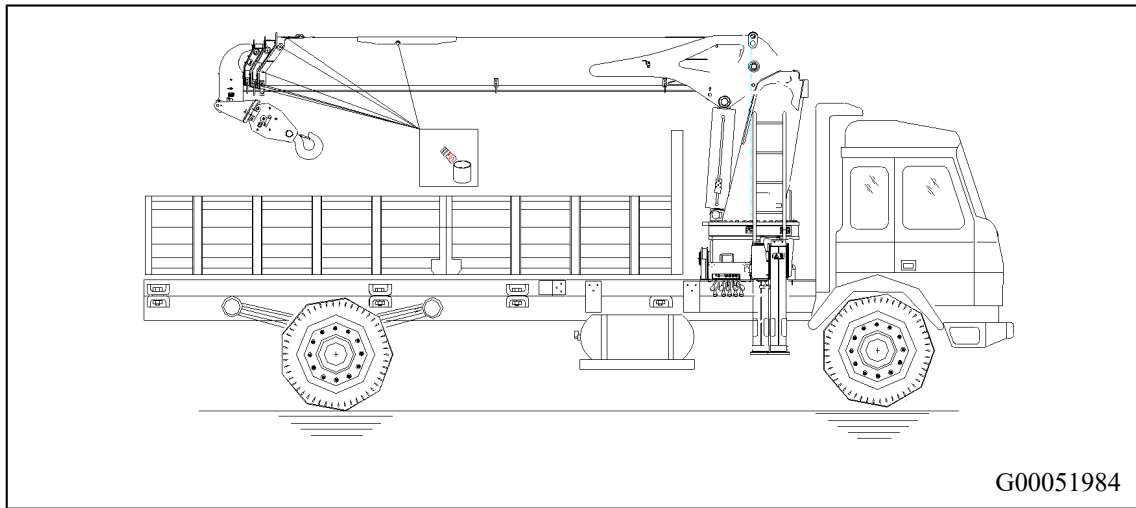
Lubrication of knuckle point

Periodic lubrication shall be applied to each knuckle point and oil cup of sliding block as shown in the figure. Inject grease at certain pressure until grease spills between two connecting pieces. After fill oil, operate each knuckle point to perform intact action for several times and then add a little more grease.



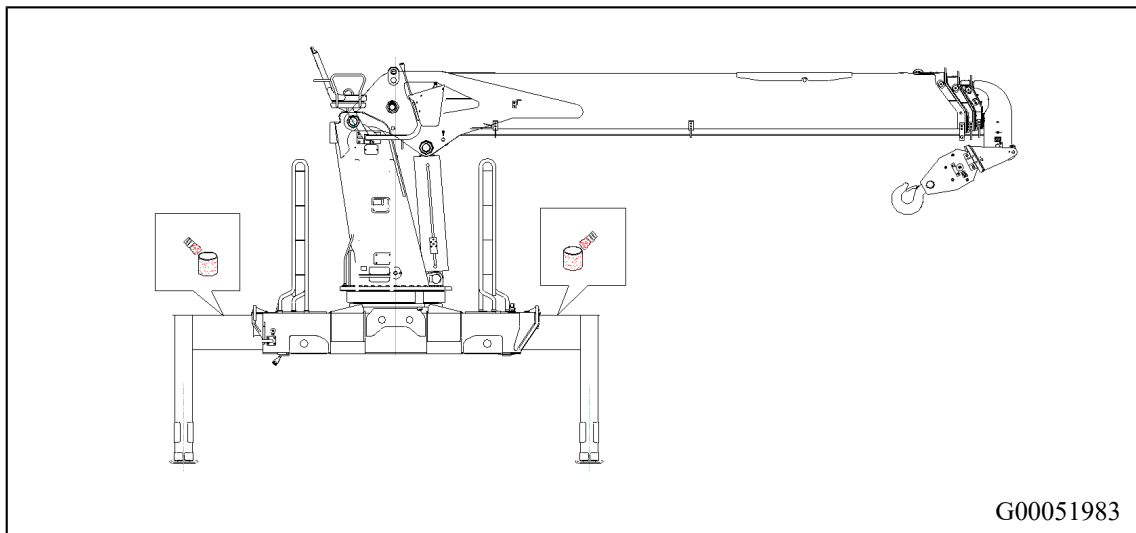
Lubrication of boom extension

Parking crane at the position as shown in figure and lubricate outer side of telescopic boom. Scrape original grease with rubber knife before applying new grease and coat a layer of new grease with brush on entire external surface of telescopic boom.



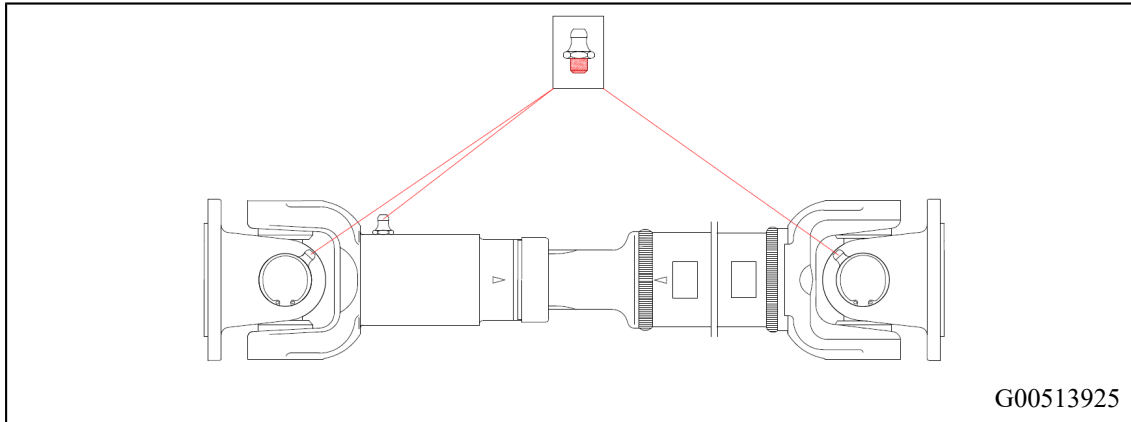
Lubrication of stabilizer extensions

Park crane at the location as shown in figure, open stabilizer extensions completely, coat a layer of new grease with brush on entire external surface of stabilizer extensions.



Lubrication of transmission shaft

Periodic lubrication shall be applied to transmission shaft splines and universal joints as shown in the figure, inject grease at certain pressure until grease spills between two connecting pieces.

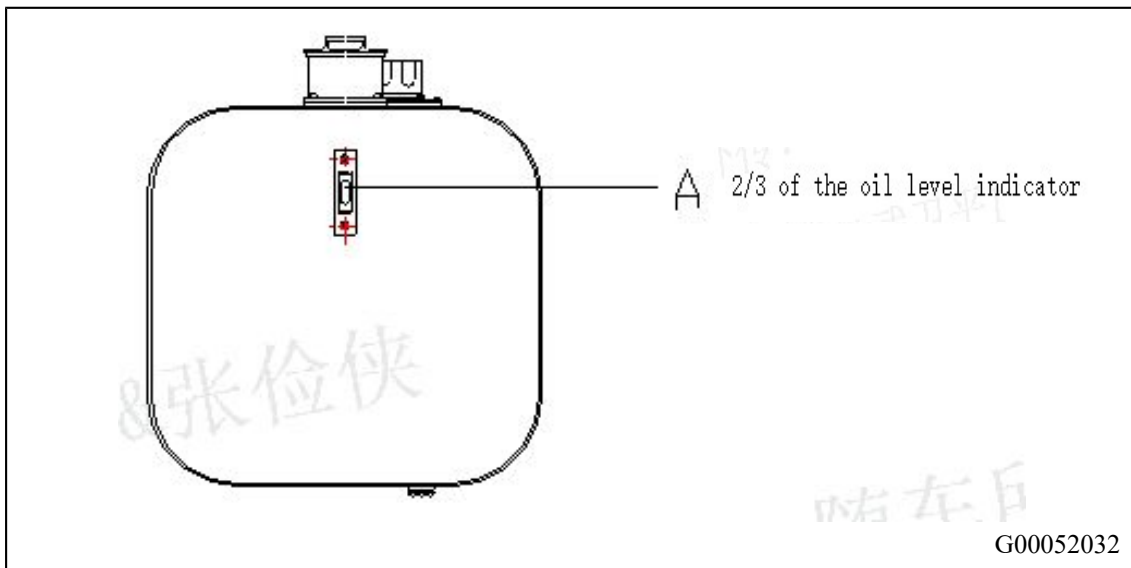

CAUTION

Always use the grease with the same characteristics as those specified in this section.

4.4 Check the oil level in the tank

Park and set the crane at close status completely, check on the level indicator that the oil reaches the position marked “A” in the illustration.

A: Maximum oil level

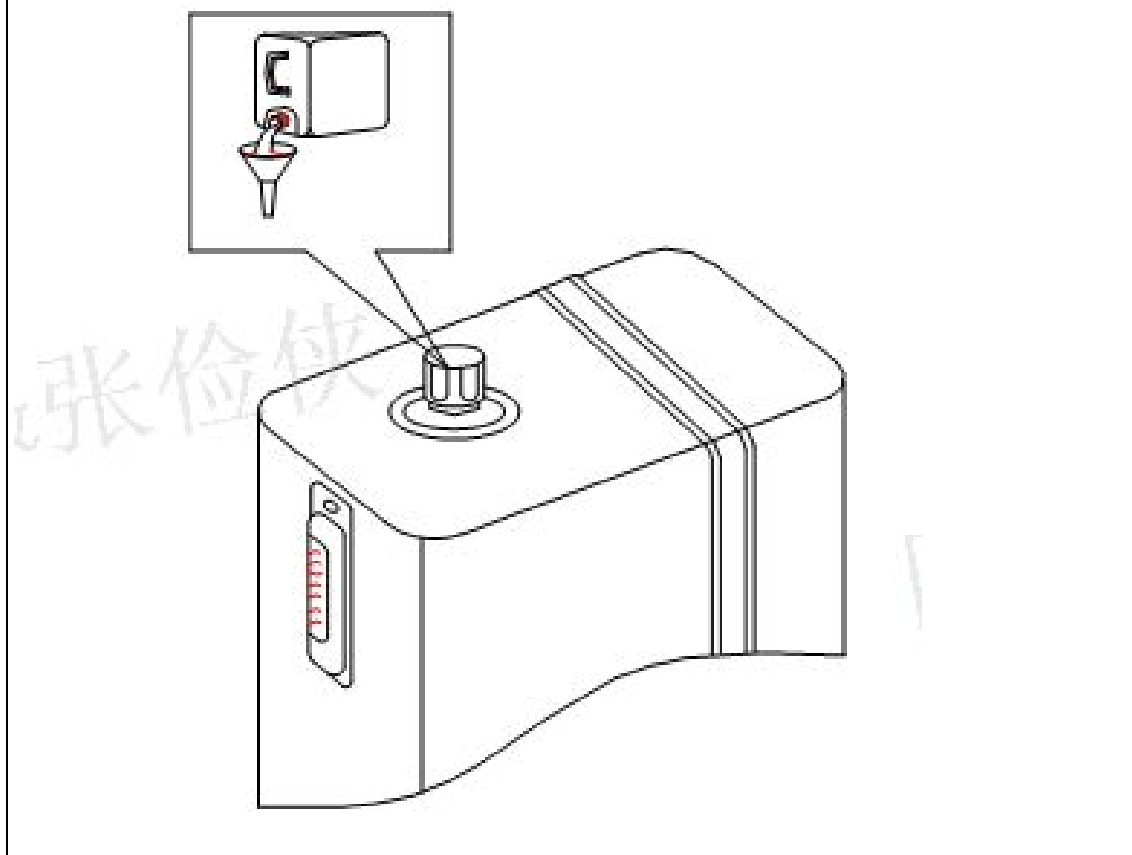


4.5 Replace hydraulic oil

Configure the crane at close status completely, find a container big enough to fill the oil and place it under the tank, then open the plug in the bottom of the tank to exhaust the oil, recover the plug and make sure that it is tight.

⚠ CAUTION

Take suitable precautions to prevent the risk of burns due to contact with hot oil.

**⚠ WARNING**

Always use oil with the same characteristics as those specified in Section “H.11”. inject the fresh oil into the tank with filter cartridge, then recover the tank cap.

4.6 Replace the oil filter

The oil filter must be replaced at the specified intervals and whenever the optical indicator shows that the filter is clogged.

Open the oil filter cover and take out filter, Replace the filter with one that has the same filtration mesh, clean it with a nitrate-based solvent, grease and check the position of seal between cover and filter body, then recover the filter cover .

⚠ CAUTION

Take suitable precautions to prevent the risk of burns due to contact with hot oil.



4.7 Check wear on sliding pads

Be sure to check the wear on the sliding pads between the boom extensions at the specified intervals. Excessive wear on the pads may cause the friction among the metal parts, the non-sequential opening of the boom extensions and/or excessive loose may leading to misalignment of the various extensions.

4.8 Check screw tightness

Vibrations may cause the screws loose which securing the crane components to work. Be sure to check the screw tightness below at the specified intervals.

A- fastening plate screws B- Boom extension nuts C-Screws and nuts on control levers D-Distributor support screws

4.9 Operation and maintenance for lifting winch

4.9.1 Purpose and character of mechanism

Lifting winch is used for fixing goods, therefore it is an absolutely necessary part for loader crane and one of the most important basic organizations of the loader crane .

The single stage NGW planetary gear transmission is adopted in the design of the lifting winch, three planetary wheels are designed on the planetary frame to achieve the purpose of uniform load, so this mechanism has characteristics of high efficient , small volume , light weight , low voice, compact structure , steady running, big scope transfer power etc.

Brake friction slice: wet type friction slice is placed in the brake shell, the brake shell is installed on the front of organization , convenient for replacement of friction slice.

4.9.2 Reasonable use and lubrication of mechanism

Do not use lifting winch to lift person !

Do not put the lifting winch in the place containing acid or alkaline etc corroded gas, make sure that the actual load does not exceed the rated load. To ensure safety, leave at least 3~6 winds of wire rope on the winch drum , and make sure that the wire rope is regularly wound on the winch drum, if you found disorderly-wound condition, be sure to arrange it again under the professional 's guidance.

The normal working temperature of lifting winch is : -25°C~+80°C, the ambient temperature is: -25°C~+40°C.

Using abrasion-resistant hydraulic oil, replace hydraulic oil at the regular intervals, replace the oil within 30 days after the organization is delivered, thereafter replace it every 3~6 months

Model of lubricant oil:

summer character	winter character
L-HM46	L-HM32



4.9.3 Storage

If you don't use the lifting winch for a long time (more than 3 months), be sure to drain the oil from lifting winch, and leave it full of turbine oil with lower acidic value;

Put it in dry place without odour of decaying ; To avoid quickening the ageing process, do not store it for a long time in the place where the temperature is too high or the temperature is below -25°C, drain out the lower acidic value turbine oil and add proper brand of grease when you use the machine again.

4.9.4 Faults and eliminations

No.	Faults	Causation	Elimination
1	Drum doesn't work while start motor	Improper installation of motor	Check the joint between motor and speed reducer(for example, whether the motor keys are installed or not)
		Brake doesn't work	Check brake system
		Internal failure	Contact our company
2	Brake doesn't work	If the brake is equipped with adjusting bolts	Turn the four bolts with clockwise
		Residual pressure in the loop	Check hydraulic loop
		Friction slice wore down	Check and replace friction slice
3	Brake doesn't release	Brake doesn't work because of being kept for a long time	Pressure is applied to run the brake
		No pressure in the brake	1. Check external connection of brake 2. Replace seals
4	Vibration or noise is big	Improper installation of lifting winch	1. Check if mounting screws are loose 2. Uneven mounting surface needs to be corrected or replaced
		Internal failure	Contact our company
5	Mechanism overheated	Lack oil	Add oil

No.	Faults	Causation	Elimination
		Brake is not fully released	Check brake, turn the four bolts with anti-clockwise, adjust brake pressure.

4.10 Operation and maintenance of slewing reducer

4.10.1 Purpose and characters of mechanism

Slewing mechanism is used for fixing goods and loader crane , therefore it is an absolutely necessary part for loader crane, is also one of the most important basic organizations of the loader crane.

The conventional NGW planetary gear transmission is adopted in the design of the slewing mechanism, which has the characteristics of high efficient , small volume , light weight , low voice, compact structure , steady running, big scope transfer power etc.

Using dry type friction plate , which is placed in the brake housing, the brake housing is installed on the upper end of the mechanism, easy to replace friction plate.

4.10.2 Reasonable use and greasing

Before the first use of slewing mechanism after installation, check if the mechanism is installed according to the installation requirements.

Before operation of the slewing mechanism, check if the pins of fixing column have not been pulled out. When travelling, check if the pins of fixing column are inserted to prevent slewing reducer from impact. Slewing mechanism shall not be stored in the environment containing corrosive gas such as acid and alkali, and not exceed its rated load during operation.

The normal working temperature of slewing mechanism is : $-25^{\circ}\text{C}\sim+80^{\circ}\text{C}$, the ambient temperature is: $-25^{\circ}\text{C}\sim+40^{\circ}\text{C}$.

Using L-CKD220 gear oil, replace lubrication oil at the regular intervals, replace the oil within 30 days after the slewing organization is delivered, thereafter replace it every 3~6 months

4.10.3 Storage

If the slewing mechanism is not used for a long time (more than 3 months), make sure that the oil in the slewing mechanism should be drained away, and fill it with turbine oil with low acid value, and store it in a dry, non-corrosive gas environment, do not store it for a long time where the temperature is too high or the temperature is below -25°C , so as not to quicken the aging of seals. When it is used again, drain the turbine oil and add the corresponding brand lubricating oil.



4.10.4 Common faults and eliminations

No.	Faults	Causation	Elimination
1	Output gear doesn't turn while motor start	Improper installation of motor	Check the joint between motor and reducer(for example, whether the motor keys are installed or not)
		Brake doesn't work	Check brake system
		Internal failure	Contact our company
2	Brake doesn't work	Residual pressure in the loop	Check hydraulic loop
		Friction slice wore down	Check and replace friction plate
3	Brake doesn't release	Brake doesn't work because of being kept for a long time	Pressure is applied to run the brake
		No pressure in the brake	1. Check external connection of brake 2. Replace seals
4	Vibration or noise is big	Improper installation of gear reducer	1. Check if mounting screws are loose 2. Uneven mounting surface needs to be corrected or replaced
		Internal failure	Contact our company
5	Mechanism overheated	Lack oil	Add oil
		Brake is not fully released	Check brake pressure.

4.11 Hydraulic oil, grease and gear oil characters

You shall estimate whether you need to install a equipment to lower the temperature or keep hydraulic oil at proper temperature in light of climate and general working conditions.

Designation of hydraulic oil:

summer Winter

YC-N46 YC-N32

Grease: 2# lithium-base grease

Model of gear oil : HL-20 in summer HL-30 (SY1103-77) in winter

⚠ WARNING

In general, when you add or replace hydraulic oil, the new oil shall be compatible with oil in hydraulic pipeline. Prohibit use of oil containing molybdenum and sulphide, since these will damage to supporting mechanism.

4.12 Disconnection

Be sure to contact manufacturer if you want to take down the machine

⚠ WARNING

Do not take down valve and locks.

4.13 Environment protection

During operation, maintenance and repair of the crane, be sure to safe use and dispose of waste in accordance with the requirements of national environmental and occupational health laws and regulations.

To protect the environment on which we live, please properly disposal of the waste during operation and maintenance of the product .

The waste should be collected, stored and disposed separately during operation, maintenance and repair of the product.

1、 The bucket for collecting Solid waste (waste paper, waste metal) is made specially according to certain requirements, recycled by recycling company.

2、 Danger waste: ① waste oil: Collecting all sorts of waste oil with special bucket and marked “danger waste” on the bucket, try to avoid leakage when collecting the waste oil. Finally send it to a qualified unit for recycling . ② Storage battery: Send the battery to the hazardous material storage place and attach the sign marked “danger waste” , after that send it to a qualified unit for recycling . +



Chapter 5 Operator

5.1 Essential requirements

Any person using or operating the crane must be competent to do so and must satisfy the following requirements:

Physical

Good eyesight, hearing, coordination ability and can safely perform all the tasks required to operate the crane. He must not be affected by disorders or disabilities and his faculties must not be impaired by the taking of drugs and/or alcohol.

Mental

Ability to understand and apply the rules, regulations and safety precautions. He must be alert and use his judgment to ensure his own safety and that of others. He must want to do the job correctly with mental condition.

Emotional

He must be calm, able to withstand stress and use his judgment to assess his physical and mental condition.

Training

He must have read, studied and understood this manual, as well as the charts and diagrams and the warning plates and notices. Prior to any lifting operation with the crane, he must be familiar with the use of the controls and the associated movements, and must have carried out no—load operation before applying a load.

He must have a license(if required by law).

Recommended clothing

Safety helmet, gloves, accident-prevention shoes, protective goggles.

5.2 Duties and responsibilities

Duties

Crane operator has the duty to master the operation of crane, ranging from start-up of power system and operation of the assembly/disassembly of the optional extras and maintenance work.

The operator must therefore be fully aware of the safety rules and devices and must also be familiar with the use of the operating controls and load harnessing techniques.

Responsibilities

The operator is directly responsible for the operation and correct functioning of the crane, for its maintenance, for the load being handled and for every action he performs in the context of the duties referred to above.

5.3 Control stations

The illustrations on this page show the positions(right and left side of the machine) which the operator must occupy when operating the crane.

CAUTION

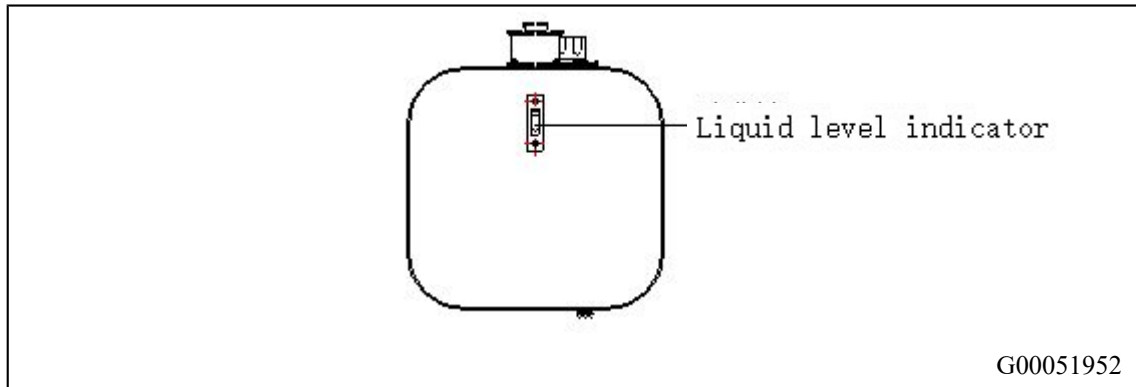
Remember that you must **ALWAYS** use the controls on the same side as the load, adopting a natural and well-balanced operation style.



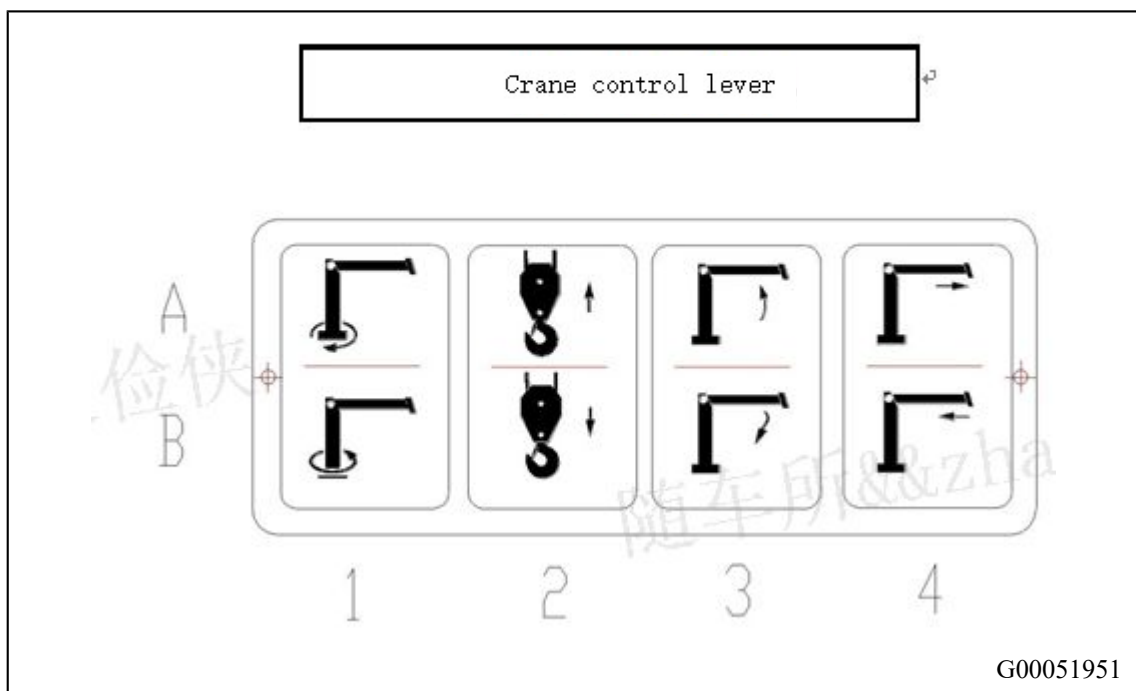
Chapter 6 Control devices

6.1 Hydraulic oil level indicator

The crane's oil tank is fitted with an optical level indicator for checking the amount of oil in the tank.



6.2 Operation and control



1— column slewing control lever

Position A: rotation with clockwise

Position B: rotation with anti-clockwise

2— wire rope control lever

Position A: up

Position B: down

3— boom luffing control lever

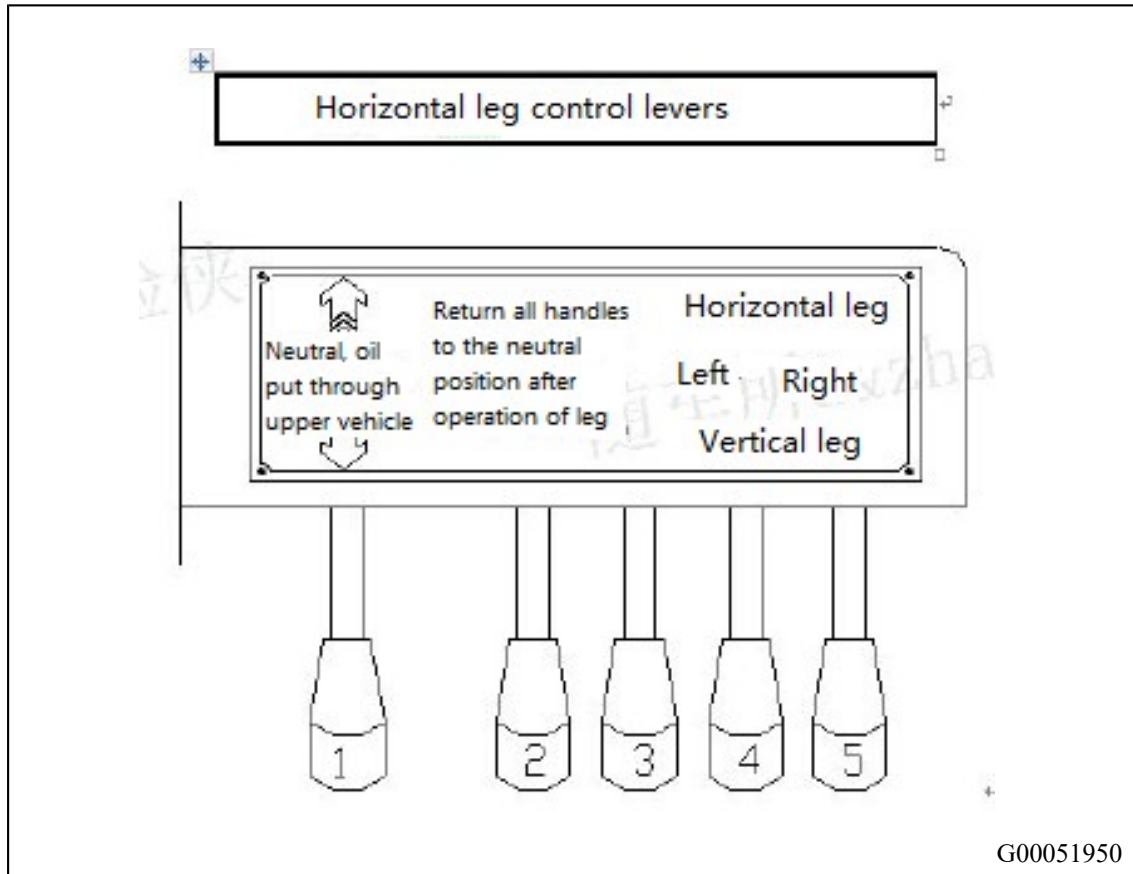
Position A: up

Position B: down

4— boom extension control lever

Position A: extend

Position B: retract



Pull the control lever 2,3,4 and 5 to the upside: crane legs and rear legs extend horizontally if pull the control lever 1 upward.

Pull the control lever 1 downward, crane legs and rear legs will retract horizontally.

Pull the control lever 2,3,4 and 5 to the downside: crane legs and rear legs extend vertically if pull the control lever 1 upward.

Pull the control lever 1 downward, crane legs and rear legs will retract vertically.

⚠ CAUTION

Return all the control levers to their neutral position after horizontal and vertical legs are extended completely.



Chapter 7 Operation instruction

7.1 Notice before operation

DANGER

- 1、 Do not use the crane until the stabilizer is opened and correctly locked in place.
- 2、 Before connecting PTO, you should check if all levers are in their neutral position. return all control levers to their neutral position.
- 3、 Open the cut-off valve on the hydraulic oil tank before start crane, otherwise the pump will be damaged.

WARNING

Before starting work with the crane, make sure that truck has enough fuel to avoid a forced stop during a manoeuvre and check that the battery is in good condition so that you can restart the truck if a forced stop occur.

CAUTION

Before using the crane, always check if the functions of safety device on the machine are in good condition.

Do not hold the control levers as shown in the illustration marked “NO”, make sure that the lever return sharply to their neutral position when system pressure reach to the maximum.





7.2 Operation and start

CAUTION

Make sure that you have enough room to accommodate the complete opening of the stabilizer extensions and position the truck at an adequate distance to lift the load (consult the load chart).

Step the clutch and place the gearlever in neutral position, start truck, push out the hand-valve from PTO when the air pressure reach 0.5-0.6Mpa ,then release the clutch footplate slowly, the PTO will be in working status.

7.3 Crane stabilization

CAUTION

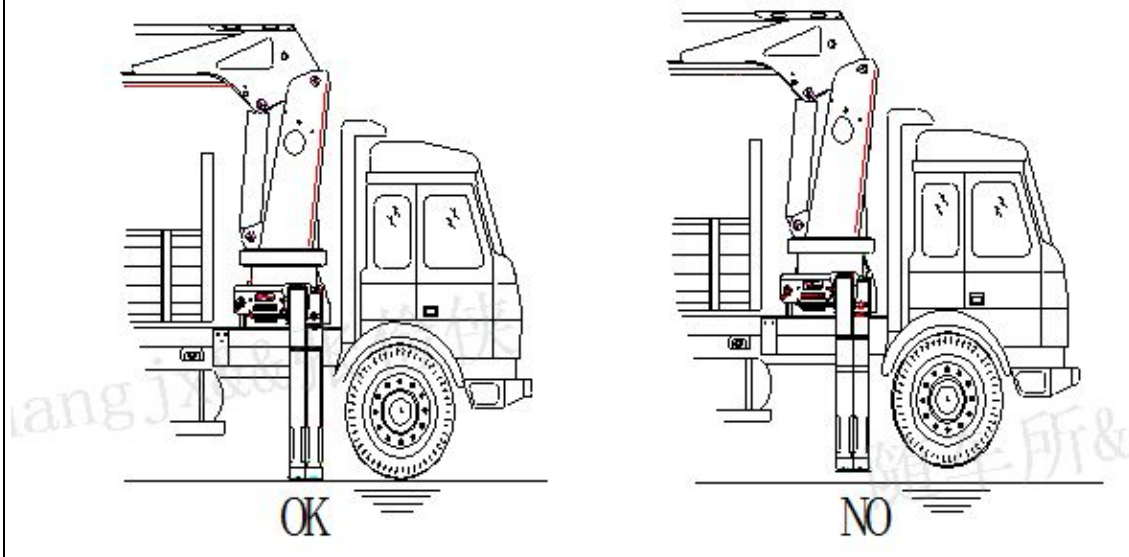
Always keep the controls located on the same side as the load you intend to perform so as to ensure maximum visibility and operator safety.

The operation of each leg can be performed as follows :

1. Unlock the locking device of stabilizer extensions.(refer to the use of lock device in safety device)
2. Pull the control lever 1,2 in the direction indicated by arrow A and B to unfold stabilizer extensions fully.
3. Lock stabilizer extensions with locking device
4. Unlock the locking device of overturning leg.
5. Pull the control lever 3,4 in the direction indicated by arrow A to extend left and right vertical cylinders to support crane .
6. Lock leg cylinders with locking device.
7. Unfold the crane.

WARNING

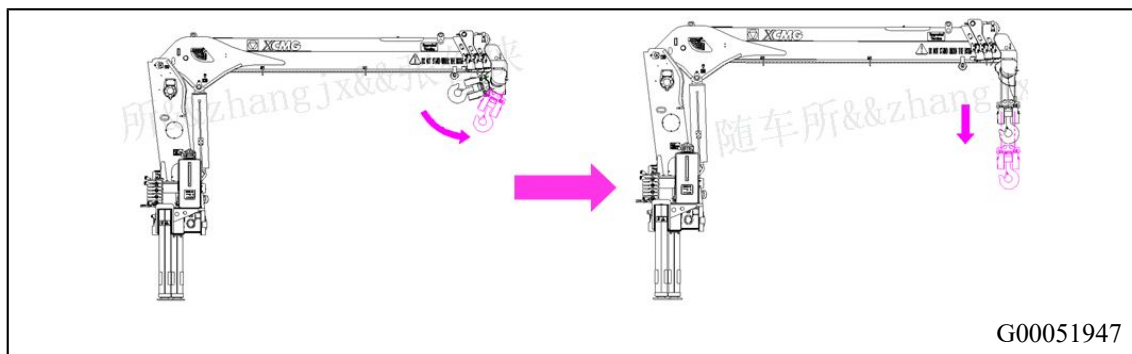
When fixing truck with loading crane, stabilizer extensions should be extended fully, the leg cylinders touch the ground and the truck shall not be suspended in the air. All wheels should be kept contact with ground .



7.4 Opening of crane

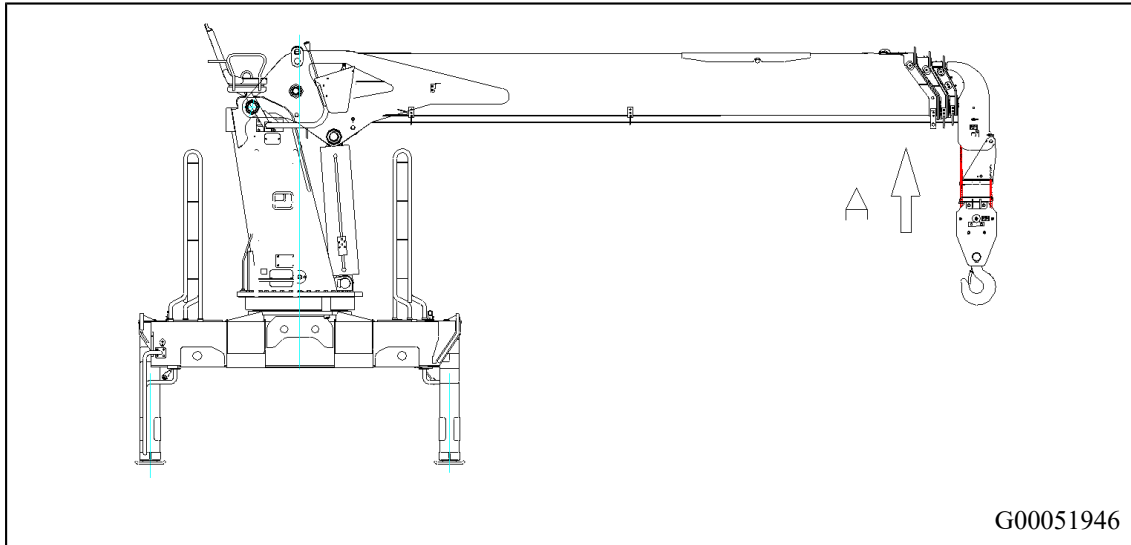
This chapter will describe all movements (actions) of crane and correct operating sequence to spread and use crane.

Before operation of crane, release automatic stowing hook mechanism and lower hook. Ensure that the hook does not rise to trigger the height limiting alarm due to luffing and extension actions.

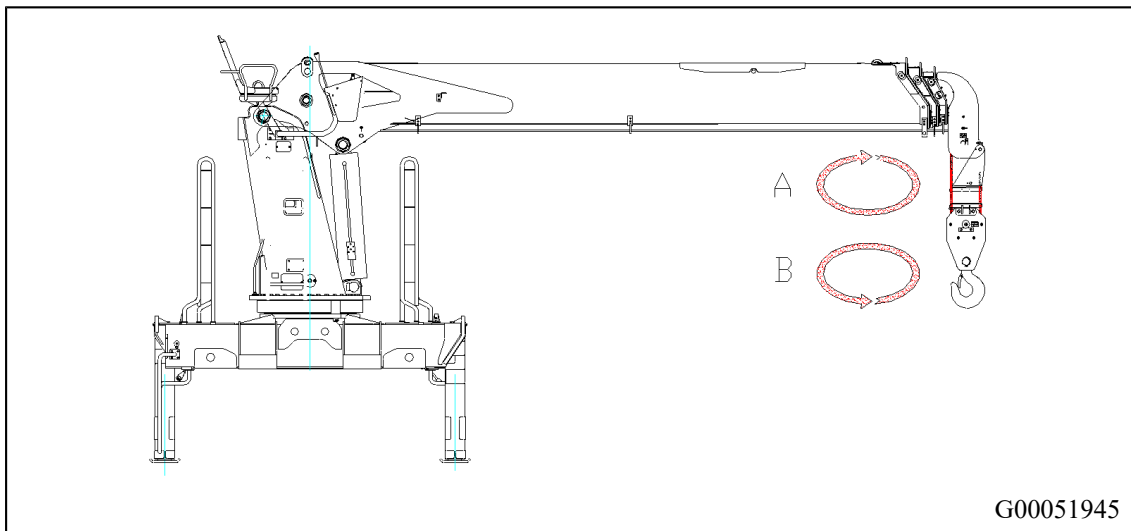


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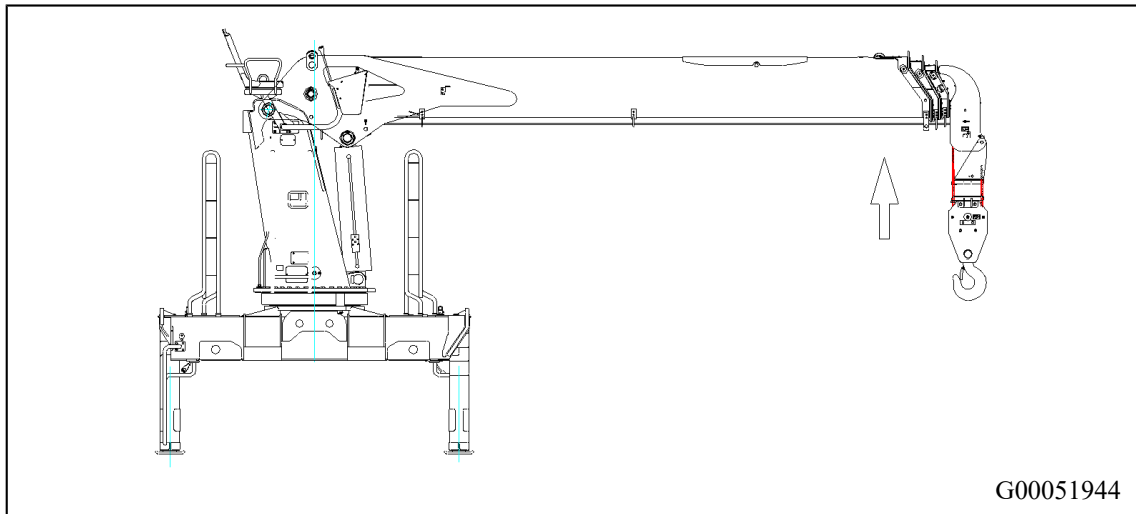
Shift control lever 3 in the direction indicated by arrow A to lift boom, on the contrary, shift it in the direction indicated by arrow B, the boom will drop.



Slewing the column with clockwise by shift lever 4 in the direction indicated by arrow A, on the contrary, shift it in the direction indicated by arrow B, the column will be slewed anti-clockwise.



Lifting mechanism: Lifting hook by shift lever 2 in the direction indicated by arrow A, on the contrary, the hook will drop by shift it in the direction indicated by arrow B.

**⚠ CAUTION**

1. When the hook is lowered, leave at least three winds of wire rope on the winch drum.
2. When the wear of the outer layer of wire rope reaches the 40% of its diameter, Or wire rope diameter is reduced by 7% or more relative to nominal diameter , the wire cable should be scrapped.

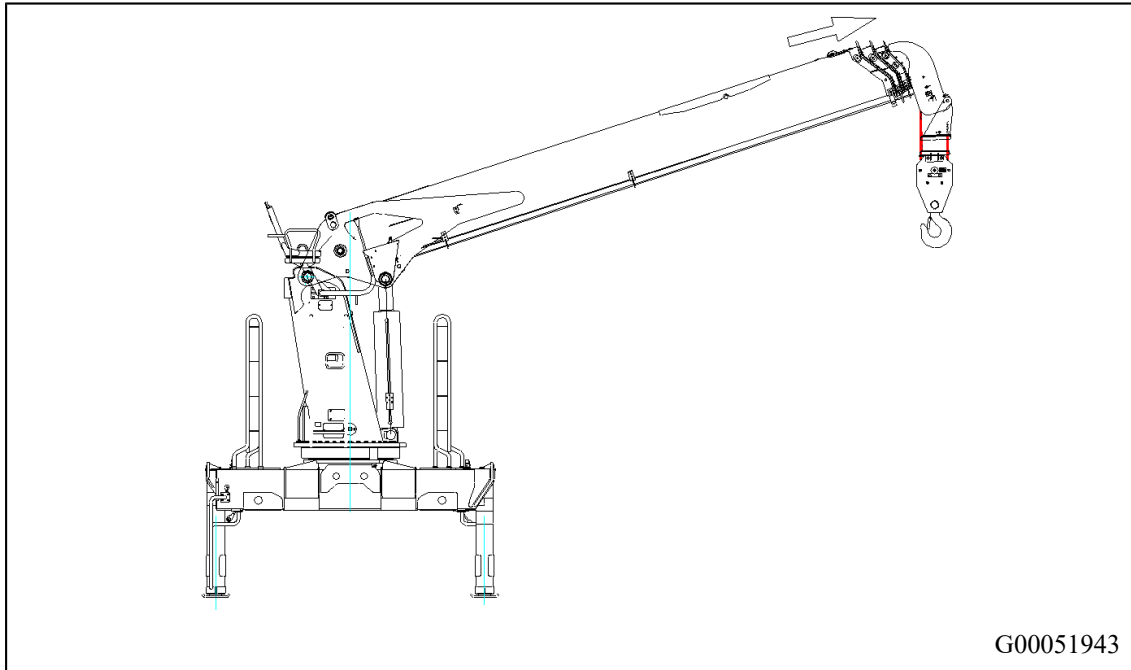
7.5 Operation of telescopic boom

Stretch boom extensions by shift control lever 1 to the position A . Before lifting goods, you must estimate goods weight (compare this value with the rated load on the lifting performance table) and work out the path of reach the required set-down point.

When you lower goods, operate slowly to make accurate landing without swing. Take care not to have goods knock crane or vehicle.

To prolong crane service life, you shall prevent column rotation with load with telescopic boom stretching. It's better to retract telescopic boom to the minimum before rotation.

After hanging goods on the hook, abrupt start and stop shall be prevented since these will generate unnecessary pressure on hoister structure.



CAUTION

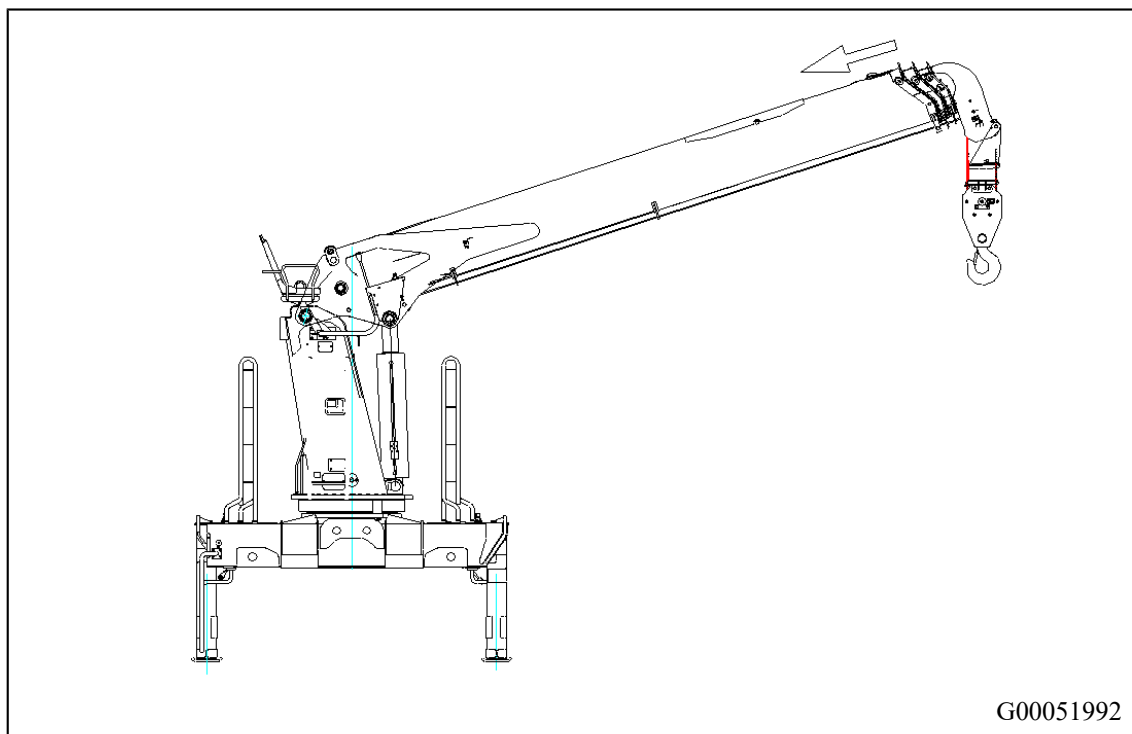
During operation of the machine, extending/retracting telescopic boom with load shall be prevented since these actions can cause excessive wear of sliding block.

7.6 Stowing crane

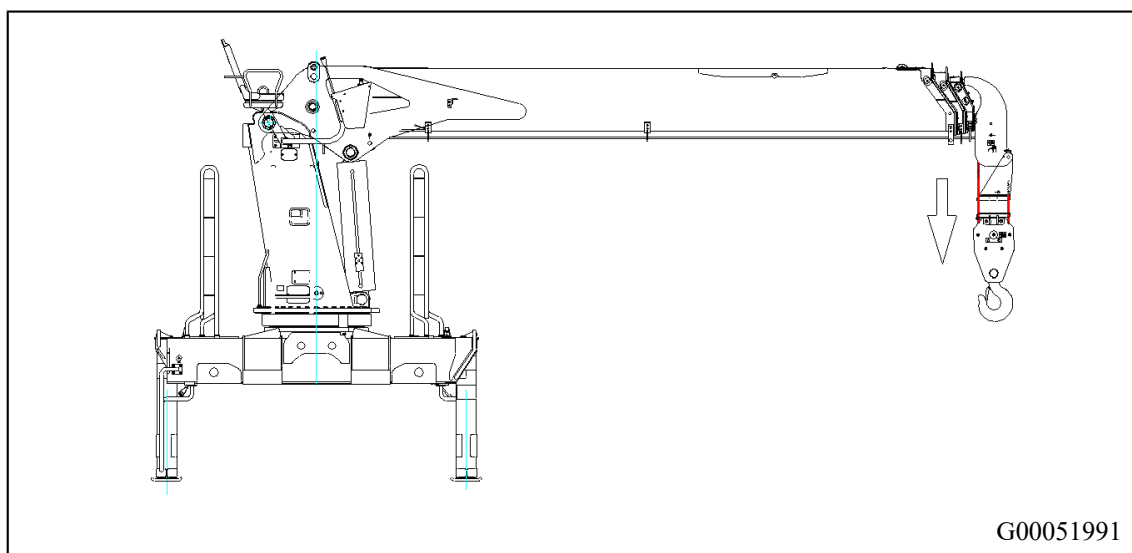
Retracting telescopic boom

Pull the control lever 1 in the direction indicated by arrow B to retract telescopic boom completely.

Caution : Do not move vehicle when the crane is in the working state.

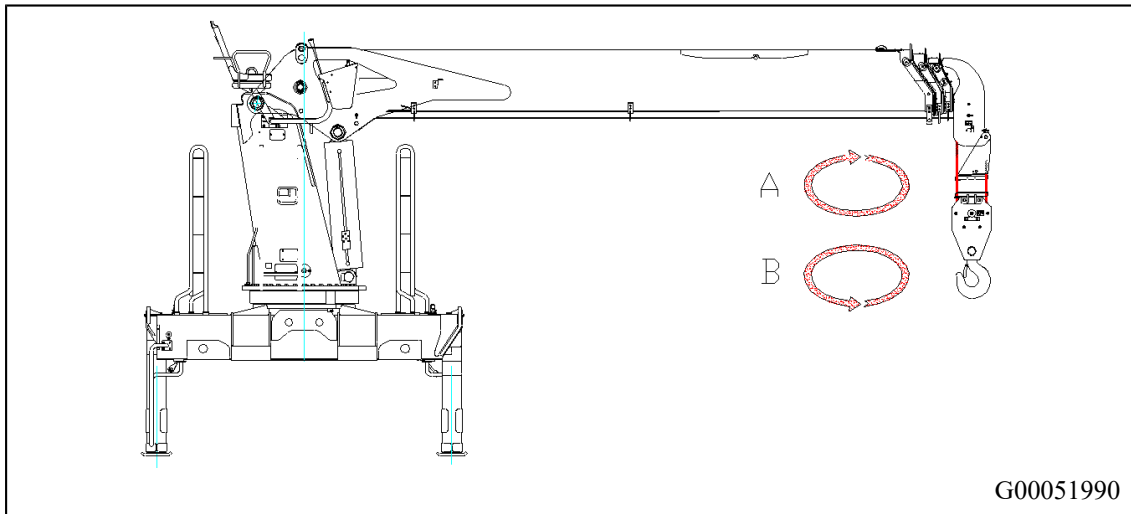


Lifting mechanism: Pull the control lever 2 in the direction indicated by arrow B to drop hook, pull it in the direction indicated by A, the hook will rise.

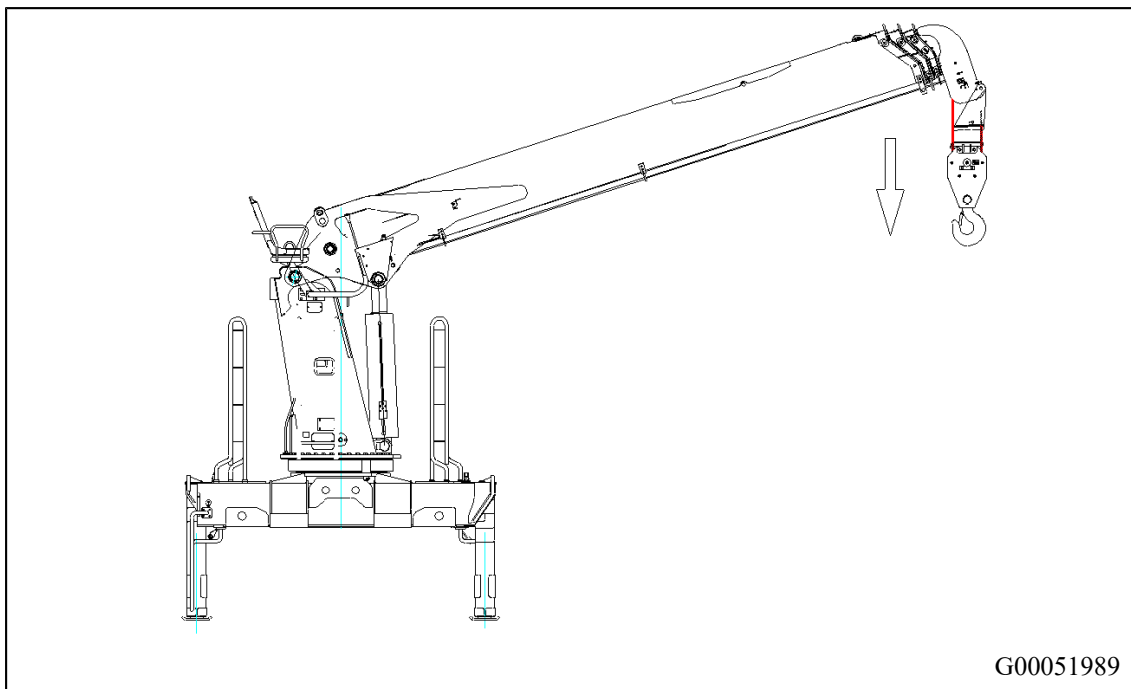


Slewing column: Slew column to place boom in the center of carriage.

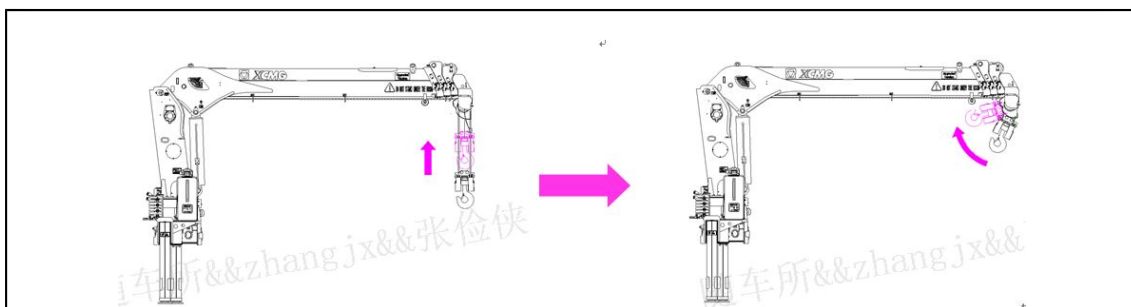
Pull the control lever 4 in the direction indicated by A to rotate column clockwise, pull it in the direction indicated by B to rotate column anticlockwise.



Stowing boom: Pull the control lever 3 in the direction indicated by A to lift boom, pull it in the direction indicated by B, the boom will drop .

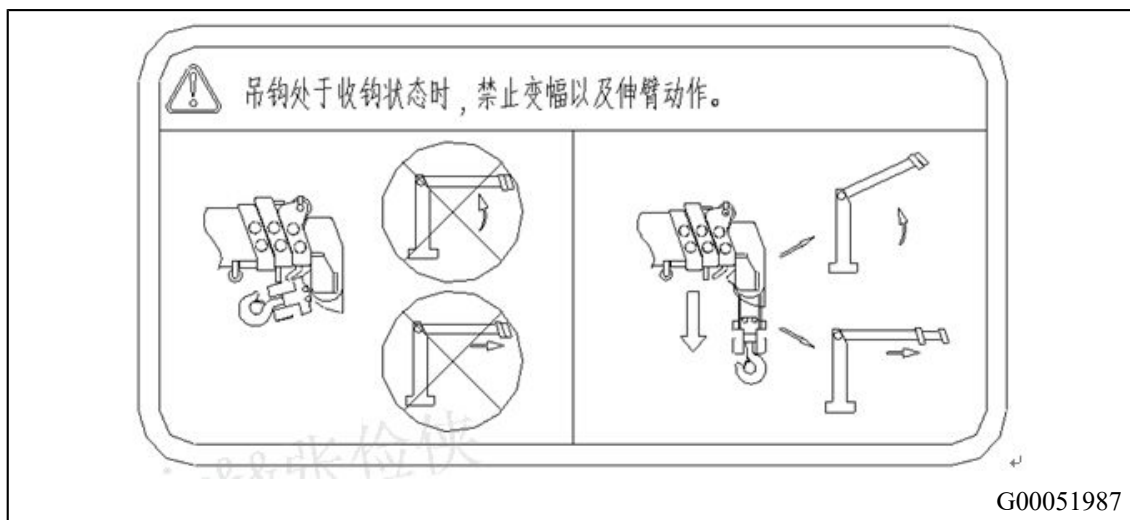


The last step of stowing crane is to tighten hook.



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Caution : After hook is stowed with automatic stowing hook mechanism, crane movements such as luffing and extension are not allowed.

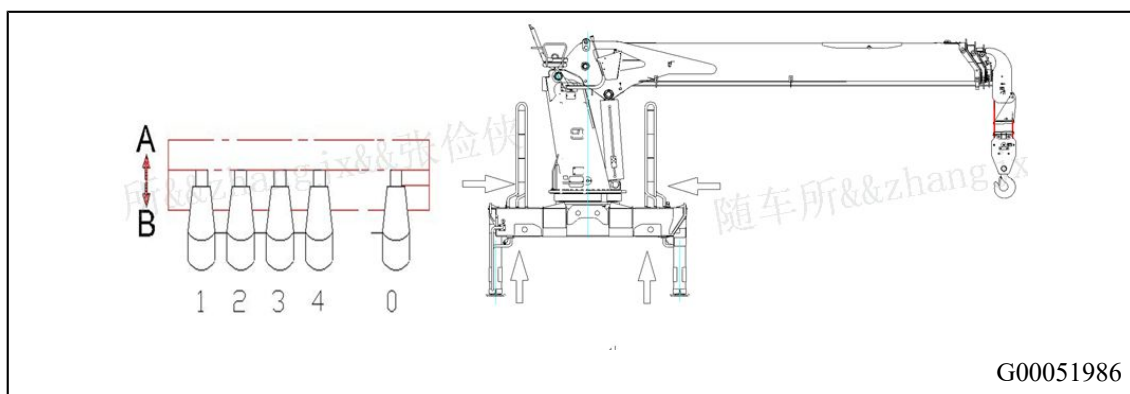


CAUTION

The stowing mechanism of crane is used to prevent hook from swing during travelling, saving the time as well. Be sure to use the automatic stowing hook mechanism when start or stop the machine. When the crane is working normally, try to avoid stowing hook to reduce the wear of pulley and raceway.

Retracting legs:

1. Pull the control lever 1, 2, 3 and 4 in the direction indicated by arrow B respectively, and pull the control lever 0 in the direction indicated by arrow B to retract left and right vertical leg cylinders.
2. Pull the control lever 1, 2, 3 and 4 in the direction indicated by arrow A, and pull the control lever 0 in the direction indicated by arrow B to retract and lock left and right horizontal cylinders.



 WARNING

Be sure to retract leg completely.

To retract stabilizer extensions, first unlock the locking lever, return it to the locked state after stabilizer extension is retracted for a few centimeters. Then retract stabilizer extension in place, and locking lever is automatically embedded and locked.

Caution : Make sure that stabilizer extensions are retract completely, and check if the locking device is engaged.

Note : 1. Divorce hydraulic pump from PTO after work is finished.

2. After operation is finished, check if control lever is automatically returned to neutral position. If not, return it to the neutral position .

7.7 Maintenance of crane

The crane should be checked regularly , check parts especially check if the function of the safety device is in well states.

1. check the connection of power line

check pipeline, all pipes & hoses should be in well condition, without any leak.

2. check oil level

check the oil amount in the tank, oil level should be between the lowest & highest signed position.

Chapter 8 Safety rules and devices

8.1 Working condition

Unless other provisions apply in the area in which you are operating the crane, if not, keep a minimum distance of 5 m from overhead power lines.

Be sure to set the machine on the firm and level ground, and the gradient is not greater than 1% and the ground surface should not sink during work.

8.2 Environment condition

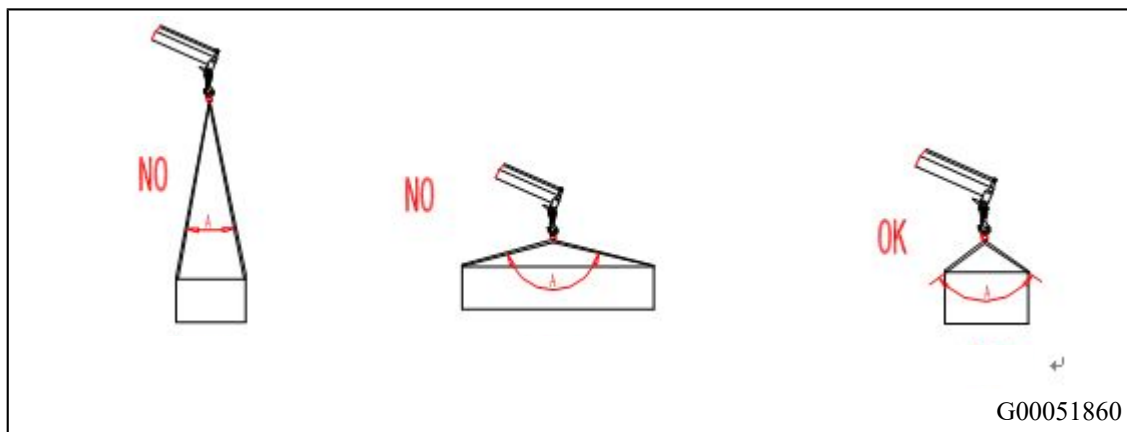
To ensure safety, before using the crane, always estimate the ambient conditions (wind, temperature, lighting).

Don't operate crane in lightning and thunderstorm weather or the wind speed is greater than 13.8m/s (grade 6).

Environment temperature shall be: -20°C to +40°C.

8.3 Requirements for hoisting wire rope

Included angle of lifting ropes (cables) shall not be too small (Figure One) or too large (Figure Two) when crane is hoisting goods. Angle A is better not exceed 120°, otherwise the lifting rope is easy to fracture due to excessive tensile force; if angle A is too small, the lifting rope will be overlong, which may result in bigger elevation angle of 1st boom during hoisting operation, and system operating pressure may be too high accordingly.



8.4 Safety rules

Make sure that operator read through this manual carefully to become sufficiently familiar with the procedure for operation, keep loader crane in the peak condition. If in doubt, read this manual carefully.

Only the person who had received training can operate the loader crane.

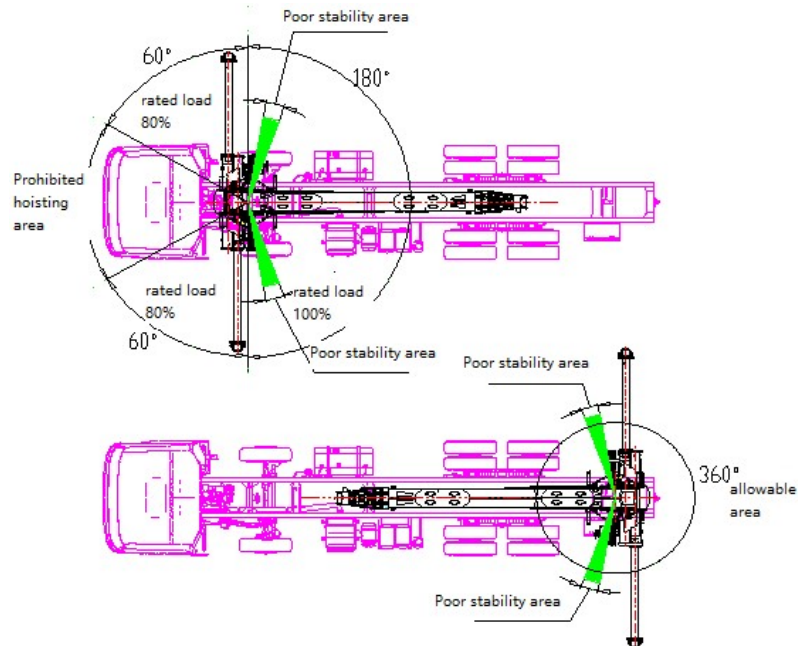
Operator shall not leave control station when lifting load or crane is running.

Make sure that no person enter the crane' s operating area without the permission of operator.

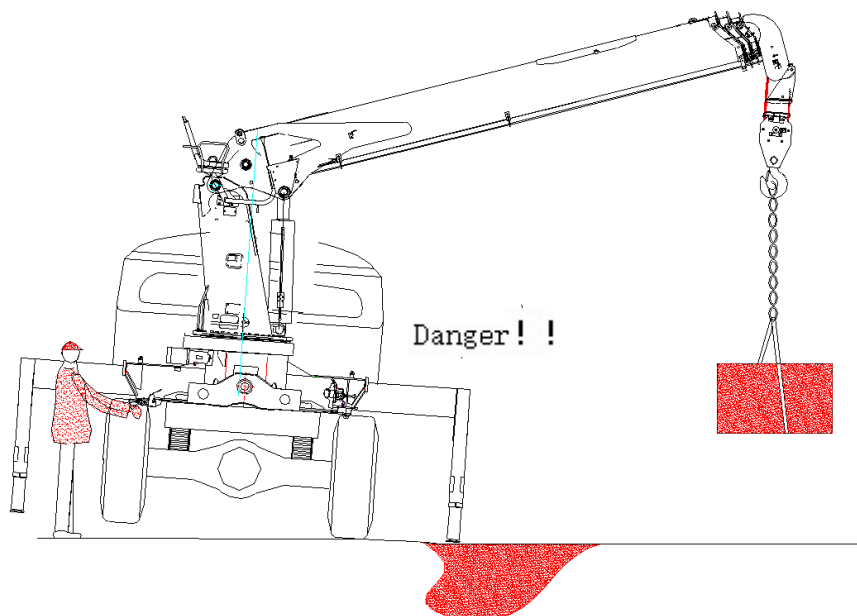
Joint operation of two types of mechanisms is strictly prohibited during hosting.

**⚠ WARNING**

Do not carry out work in the area marked with “prohibit lifting load” as shown in the figure below. Try to avoid working in the area marked “poor stability area” in the figure below. .



In order to distribute the load evenly, do not set the crane on the soft ground or ditches. When working on soft or yielding ground, you must increase the load-bearing area by placing a support plate(wood block, iron plate) between the base of the stabilizer leg and the ground. Do not operate the crane close to ditches or on soft, yielding ground.



Operator shall know the weight of goods being lifted roughly, which shall not exceed load value specified in crane performance curve.

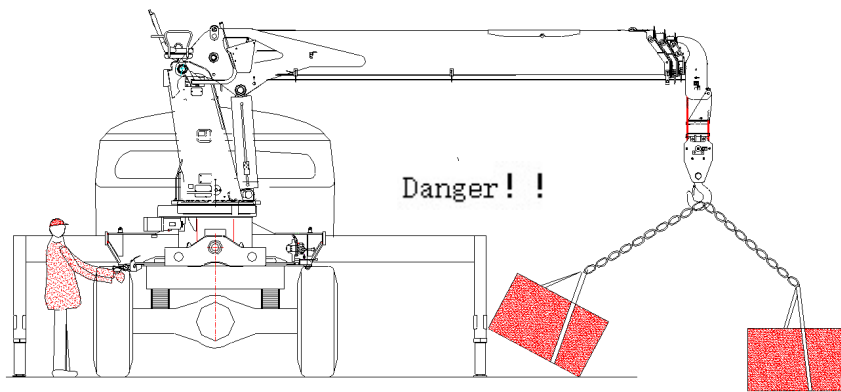
Remember to stand at the same side as goods being lifted to ensure the best visibility of load and adopt natural and normal operation mode.

Do not stand under goods being lifted.

Load shall have no angle during lifting to ensure load safe and firm.

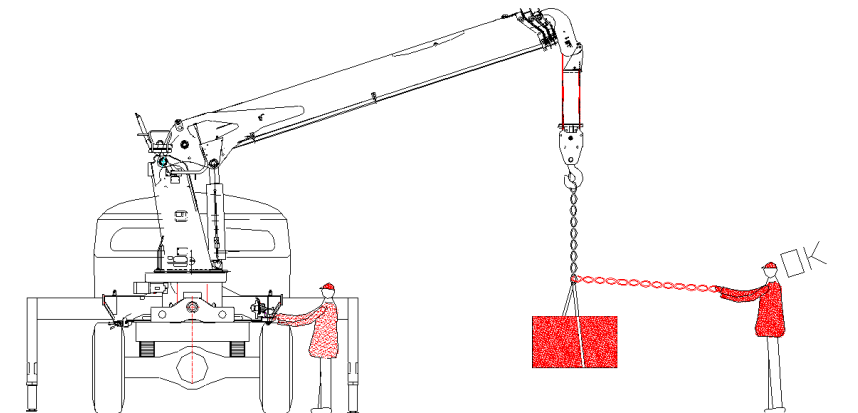
Never drag load or pull load inclined, nor lift heavy goods buried in ground or frozen to ground.

When the load just leaves the ground, make sure that there is no obstacle in the load traveling direction.

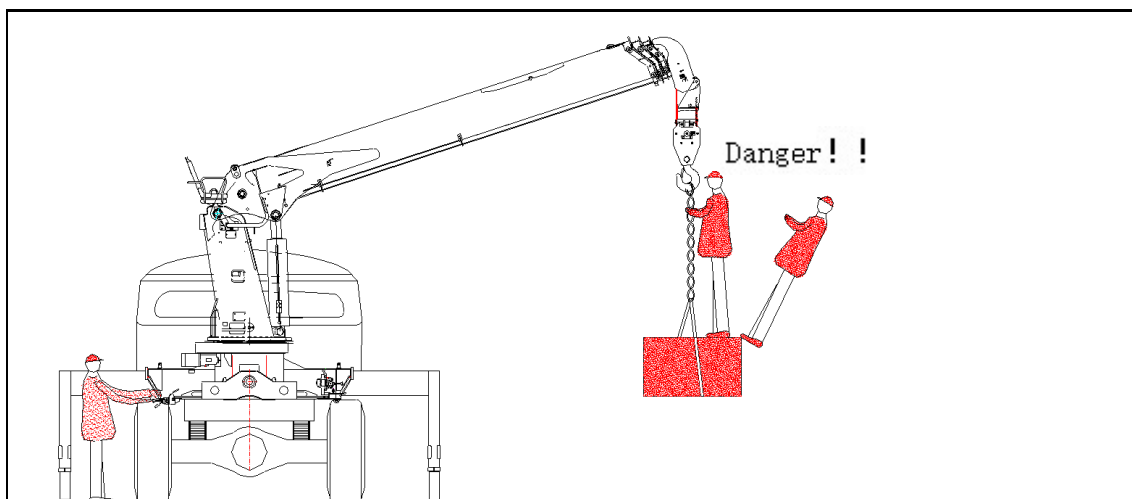


When the mass of goods being lifted is equal to or little less than the $\frac{2}{3}$ of the maximum lift capacity of crane with boom in horizontal position at corresponding operation range, the telescopic boom is allowed to retract (shorten the boom) other than stretch.

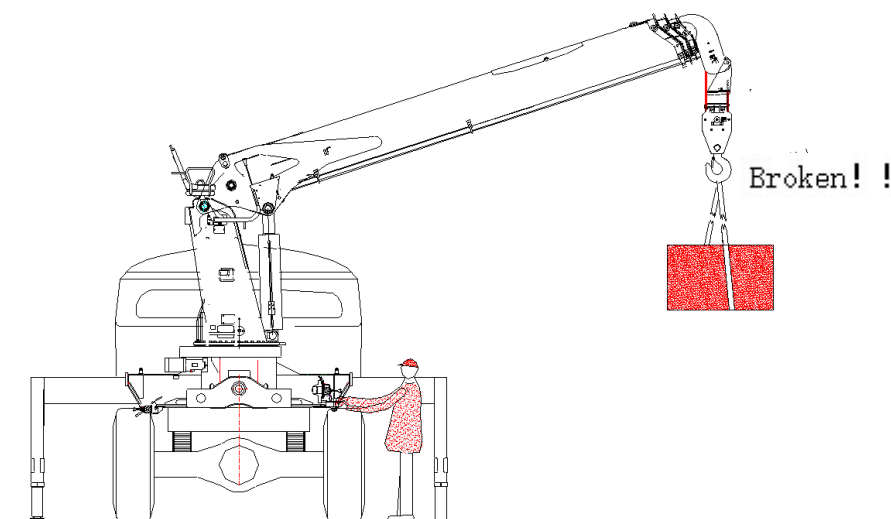
When the mass of goods being lifted is less than or equal to the $\frac{2}{3}$ of the lift capacity of crane at maximum operation range, the telescopic boom is allowed to stretch (lengthen the boom) or retract. Pilot load with rope at certain distance to prevent load from rotation.



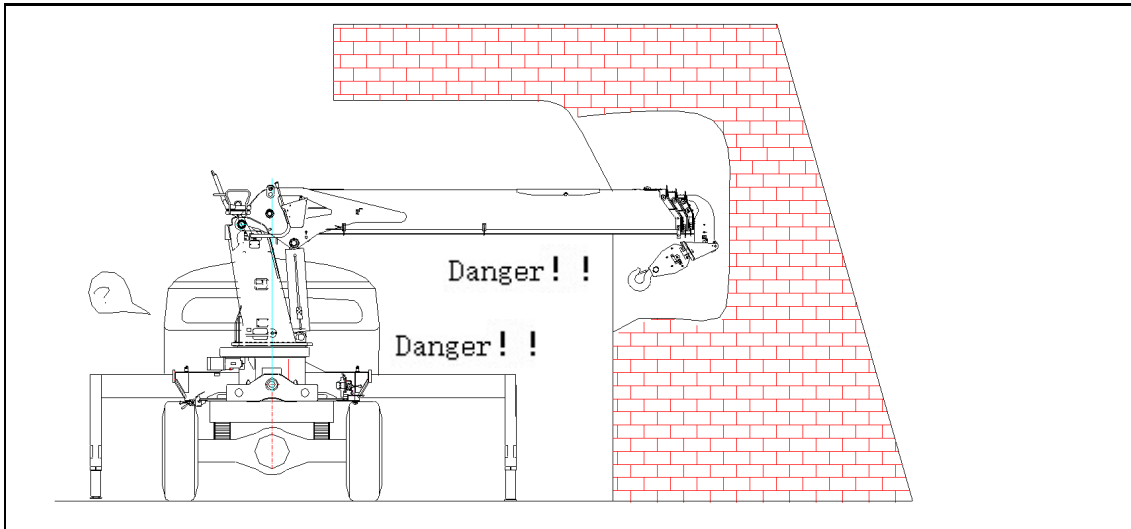
Do not use crane to lift person.



Before operation, check whether lifting appliance (rope, hook, chain or belt) has been designed safe and in accordance with requirements of goods being lifted. To ensure safety, please fix goods firmly.



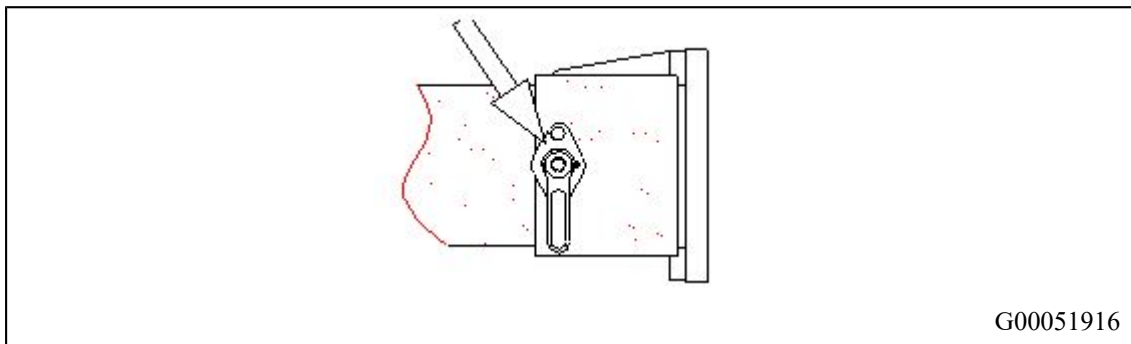
Be sure to retract and lock the crane completely before moving the vehicle.



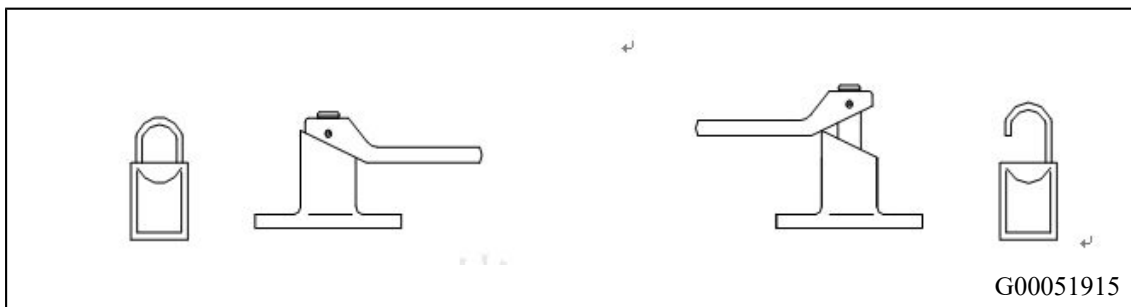
8.5 Safety devices

8.5.1 Locking devices on the fixed leg

This machine is fitted with a lock installed on the fixed legs on the right and left side of the crane.



Be sure to lock the locking device when truck is in travelling state. When extending stabilizer extension, rotate locking device of 180°, then extend stabilizer extension; rotate back locking device of 180° to retract stabilizer extension, and make the locking lever inserted in the hole of stabilizer extension.

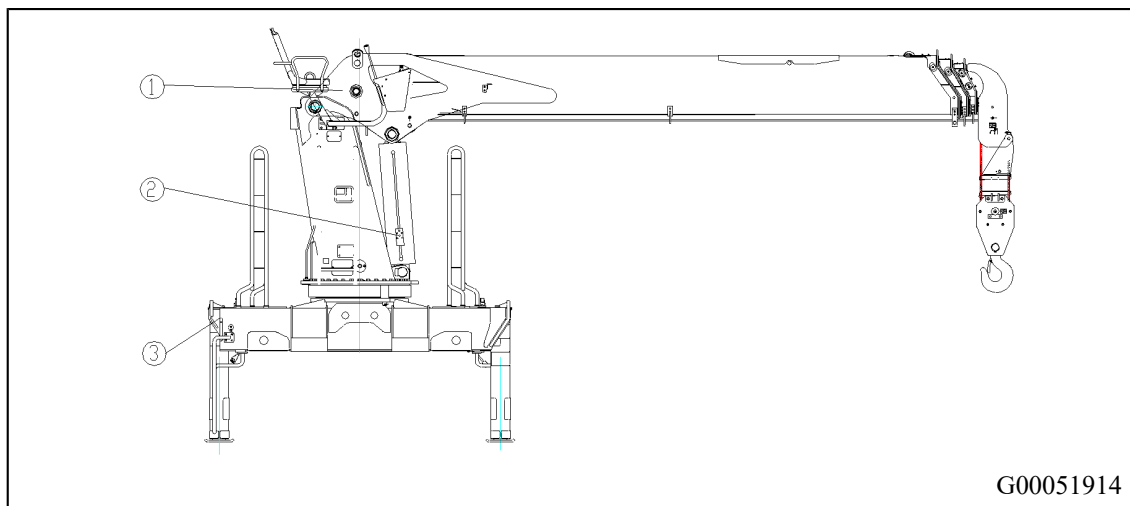


8.5.2 Valves on the hydraulic cylinder

In the case of failure in power circuit hose and insufficient engine power, the control valve still can hold load in current position and provide safety protection.

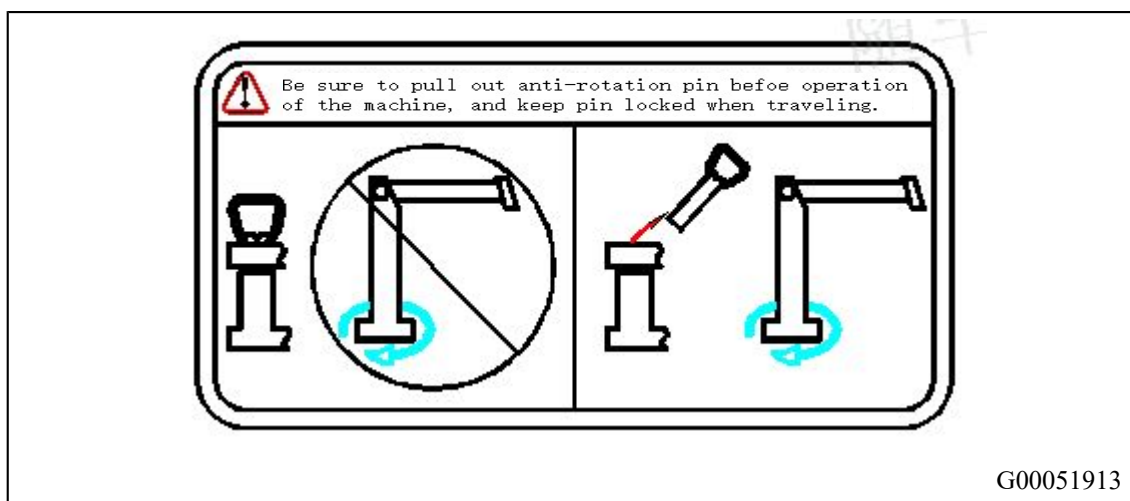
The installation positions of these valves are :

1. extension cylinder 2. luffing cylinder 3. vertical leg cylinder



8.5.3 Anti-rotation pin shaft

Be sure to refer to the warning labels on the pin shaft bush side of base when operating your machine; be sure to pull out anti-rotation pin shaft before operation of the machine, and insert it in the pin shaft hole of base after work is finished. In addition, keep pin shaft locked when travelling .



8.6 Unpredictable and predictable hazards

Unpredictable hazards: Signifies that element of risk which it was not possible to eliminated at the design stage and which is not protected by the safety devices.

Predictable hazards: Refers to any potentially hazardous point inside or close to the machine or in the crane' s operating range.

The following may be sources of hazards:

The incorrect positioning of the machine in its work configuration in which does not offer an easy escape route for the operator.

Electrocution due to contact with overhead power lines or lightning.

Asphyxia was caused by close the exhaust pipe long time.

Locking leg cylinder incorrectly(in extend or retract fully state)



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